



Southern California Edison
CS Re-Platform Project

Process 30 CIM-CS-17 Front Office

Customer Interaction Management

Business Process Design

Version 2.5

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1 Business Process Overview

As a part of the CSRP project, SCE is replacing their existing legacy system with SAP Hybris Cloud for Customer, which will be fully integrated with an S4/HANA on-premise system. SAP Hybris Cloud for Customer will include both Sales and Service functionality that will be enabled for SCE. This document will provide an overview of the key functionality that will be used by SCE.

1.1 Current Pain Points for Southern California Edison

- Navigation between two customers at the same time is not possible.
- Separate confirmation numbers are logged when a customer completes a transaction from the APA, SCE.com, and IVR (as they are all different). This is a challenge for the Call Center when the customer is calling especially for Reconnection follow-up.
- APA: there are two account number for customers with energy account and deposit – this is confusing for customer and call center for follow up. There should be a way for customer to enter the specific account number before enter payment so it's clear to know which account the payment was applied.

1.2 Responsible Parties

Contact	Contact Person
Southern California Edison Business Process Owner	Janet Covington
Southern California Edison Lead	Christine Evans
HCL Functional Lead	Adrianus Halim
Solution Architect	Pedro Costa
Front Office or Back Office Lead	Christine Evans
Subject Matter Experts	Deborah Tillman Lorena Carrasco Scott Hirth Janine Aumont Jerry Aguilar Natalie Martinez
Key User Group	Customer Contact Center (CCC) Business Customer Division (BCD) CSOD CP&S LPA Planning

1.3 Document Control

Version	Description of Change	Date	Author
0.1	Creation of Initial Version	03/09/2018	Deepak Kumar Sinha
0.2	Initial Review	03/26/2018	Adrianus Halim
0.3	Updated section 2 and 6, Modified Screenshot	03/29/2018	Deepak Kumar Sinha

Version	Description of Change	Date	Author
1.0	Updated initial draft	04/09/2018	Vasu Jasti
1.1	Updated Hybris C4C for Utilities details	04/11/2018	Deepak Kumar Sinha
1.2	Updated Review Comment and business role details	04/12/2018	Deepak Kumar Sinha
1.3	Updated section 2	04/17/2018	Deepak Kumar Sinha
1.4	Updated section 2	04/20/2018	Deepak Kumar Sinha
1.5	Updated all section based on review comment	04/24/2018	Deepak Kumar Sinha
1.6	Updated with feedback from review	04/26/2018	Vasu Jasti
1.7	Updated with feedback from review	04/27/2018	Vasu Jasti
1.8	Updated with feedback from SCE review	05/09/2018	Vasu Jasti
1.9	Modified/Updated multiple sections.	5/15/2018	D. Arebalo
2.0	Updated with feedback from SCE Review	5/18/2018	D. Arebalo
2.1	Updated with feedback from SCE review	5/20/2018	D. Arebalo
2.2	Updated with feedback from SCE review	5/22/2018	D. Arebalo
2.3	Updated FRICEW Section and added Premise section	5/30/2018	D. Arebalo
2.4	Responded to SCE comments	6/1/2018	D. Arebalo
2.5	Updated based on Credit Check Authenticate Decision	6/26/2018	D. Arebalo

1.4 Terminology

Term/Acronym	Definition and SAP Equivalent (if applicable)
APSE	Accelerated (or Axon) Project Support Environment
BCD	Business Customer Division
Billing Document	A document that outlines the charges incurred for energy consumption for a specific contract. During the process of Billing, a billing document is generated for each contract on a contract account.
BPEM	Business Process Exception Management
BMD	Business Master Data. The customer information which includes personal details, payment terms, dunning levels, billing rates and rate categories. The business master data objects are Business Partner, Business Agreement/Contract Account and Contract.
BPD	Business Process Design (or Document)
C4C	SAP Cloud for Customer Solution
CA	Contract Account
Connection Object	Link between installation and postal regional structure. Usually a building, but can also be a piece of property or other facility, such as a fountain or construction site. The connection object corresponds to the functional location in plant maintenance (PM).
CP&S	Customer Programs & Services

Term/Acronym	Definition and SAP Equivalent (if applicable)
Customer	Any entity with a relationship with SCE. This term is usually used in reference to customers (both individuals and organizations) This can include employees and third party suppliers (banks and payment partners). SAP Equivalent is Business Partner (BP)
ERP	SAP Enterprise Resource Planning
IMG	SAP Implementation Guide - where configuration is performed
TMD	SAP Technical Master Data e.g. Connection Object, Premise and Device Location
Level Pay Plans	SAP equivalent term is Budget Billing. Budget Billing allows customers to pay regular equal amounts over a yearly period.
SCE	Southern California Edison

Additional terms may be found in APSE [General Document # 32 CSRP Glossary](#)

2 Business Process Design

2.1.1 Front Office – Systems Overview

This section will provide a high level overview of the front office system architecture, and provide an understanding of the key functionality of each of the systems.

SCE is replacing their existing front office legacy systems with SAP Hybris Cloud for Customer (C4C), which will be fully integrated with an SAP S/4HANA backend system and Hybris Marketing. The front office end users will primarily be working within Hybris C4C.

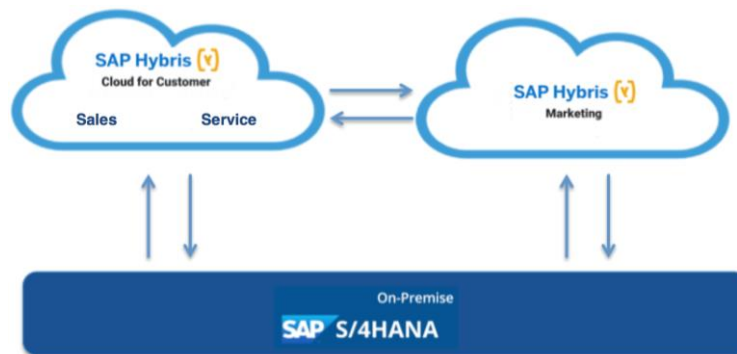


Figure 1: Overview of SAP system architecture for front office.

Hybris C4C is a cloud-based solution that can be run on a web browser. C4C can be accessed on a desktop computer, tablet, or mobile device. As long as a user is within the SCE network, the user is able to access the C4C functionalities (including front office processes, which are detailed later in this document). At the time of writing, SCE will need to further discuss how front office processes (FOP) will be enabled externally (outside the network). A user is required to have valid login credentials and the required authorization in order to access the system.

Hybris C4C has two core functional areas that will both be enabled at SCE: Hybris Cloud for Service and Hybris Cloud for Sales. An overview of each module is described below:

- Hybris Cloud for Service: This module will be used primarily by the front office users. The key functional areas include: Activities Management, Tickets management, and Utilities Call Center
- Hybris Cloud for Sales: This module will be primarily used by BCD and CP&S groups. The key functional areas include: Sales Planning, Leads Management, Opportunity Management, and Territory Management. This document will briefly discuss the sales functionality. For further information on the sales processes, please refer to [CIM-CS-16 - Manage Leads, Opportunities & Territory](#)

***Note:** Both Hybris Cloud for Sales and Hybris Cloud for Service functionality are within the Hybris Cloud for Customer system. A user's business role assignment will determine the functionality they are able to access.

2.1.2.1 UI and General Navigation - RUI vs. HTML5

In this section, the C4C UI components and navigation information will be discussed.

There are two UI's within C4C: the Responsive UI (formerly known as the Fiori UI) and the HTML5 UI. The Responsive UI (RUI) is the newer UI that has an improved 'look and feel' for end users when compared to the HTML5 UI. In addition, the design of the RUI is optimal for users that desire to use mobile tablets or touchscreens. The system administrative functions and configuration can only be done through the HTML5 UI. Therefore, the HTML5 UI will still be used by SCE, but only by users that have the elevated access to perform those functions. The default UI for all users will be set to the RUI. While all users will have the ability to switch between the two UI's, their access when they switch UI's will be the same. Therefore, a call center user that switches from the RUI to the HTML5 UI will see the layout of their screen change, but the content and their authorization will not change.

Both UI's use the same navigation elements, but they are rendered differently within each UI. The main UI elements are described below, and each area is highlighted on both UI's. An example of the two UI's rendering the same facet and navigation components are in **Figure 2 Utilities 360 Facet using the Responsive UI** and **Figure 3 Utilities 360 Facet using the HTML5 UI**

below. Since SCE will be primarily using the RUI, this document will only show screenshots of the RUI (unless the screenshot is intended to show functionality only available within HTML5).

UI Navigation Component	Description
Navigation Bar	The navigation bar is the starting point to access the functionality in C4C. The navigation bar is made up of work centers, which contain a sub-menu called work center views. The navigation bar with work centers is denoted as #1 in both UI images below. RUI: Appears vertically on the left side, which can be collapsed. HTML5: Appears across the top of the screen
Work Centers	Work areas are logically grouped within work centers. Upon selection of a work center, a submenu of functions will appear, called <i>work center</i> views. An example of this is demonstrated in Figure 2 . The work center 'Customer' was selected, and upon selection, the submenu of work center views under 'Customers' is available for selection. The work center views available are 'Accounts', Contacts, and 'Individual Customers', which are the various types of customers. Work centers are denoted as #1 in both UI images below.
Work Center View	Work center views are the sub-menu of functionality for each work center. Once a view is selected, the user can perform a specific function. This is outlined as #2 in both UI's below.
Facet	The additional tabs that appear across the top of the screen when a user is within an account are called facets (Tabs). This is denoted as #3 in both UI images below.

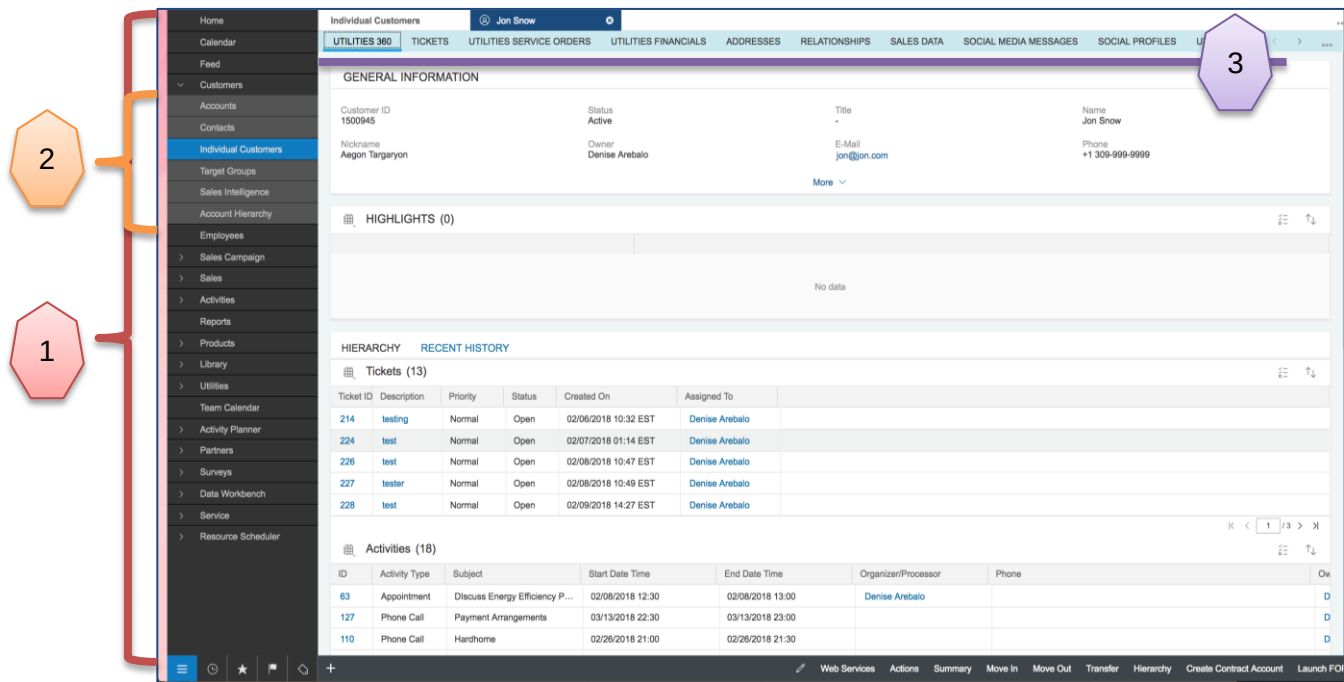


Figure 2: Utilities 360 Facet using the Responsive UI

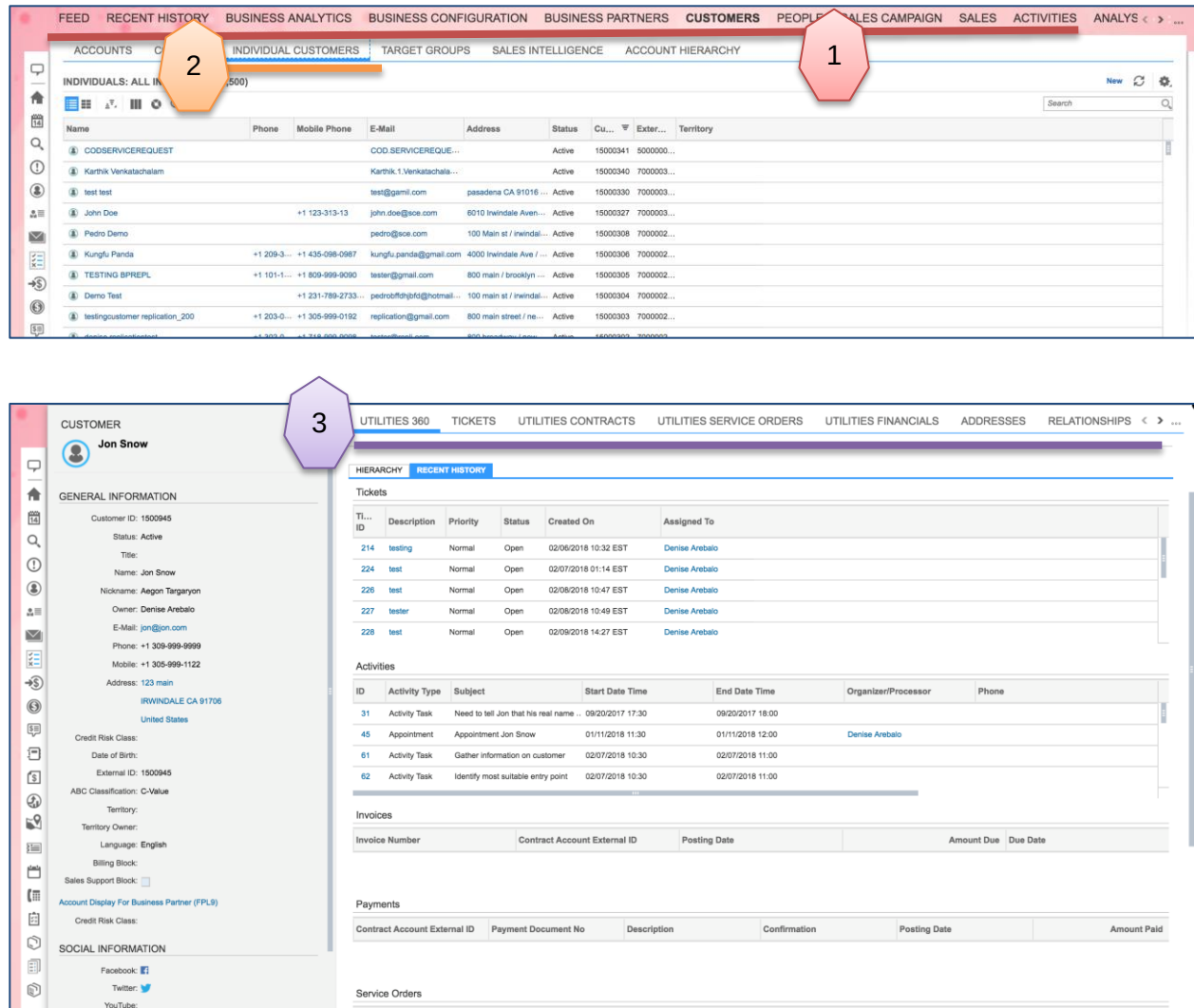


Figure 3: Utilities 360 Facet using the HTML5 UI

2.1.2.2 UI and General Navigation - Home Page

When a user logs into C4C, the landing page will be the RUI Home page. The RUI Home page provides an overview of items that require the user's attention; KPI's, reports, or dynamic Rich Site Summary (RSS) Feeds. Each tile (icon) provides an overview of the information, and serves as a link for navigation into additional details. The RUI home page can be configured for each business role to show the most pertinent information. The BCD user roles may have tiles (icons) for leads, opportunities, and sales forecasting (as shown in **Figure 4**), while a front office manager may have tiles (icons) specific to outstanding tickets. In addition to business role configuration, users also have the option to personalize their home page by adding tiles (icons) and grouping them based on their preferences. The background image can also be personalized.

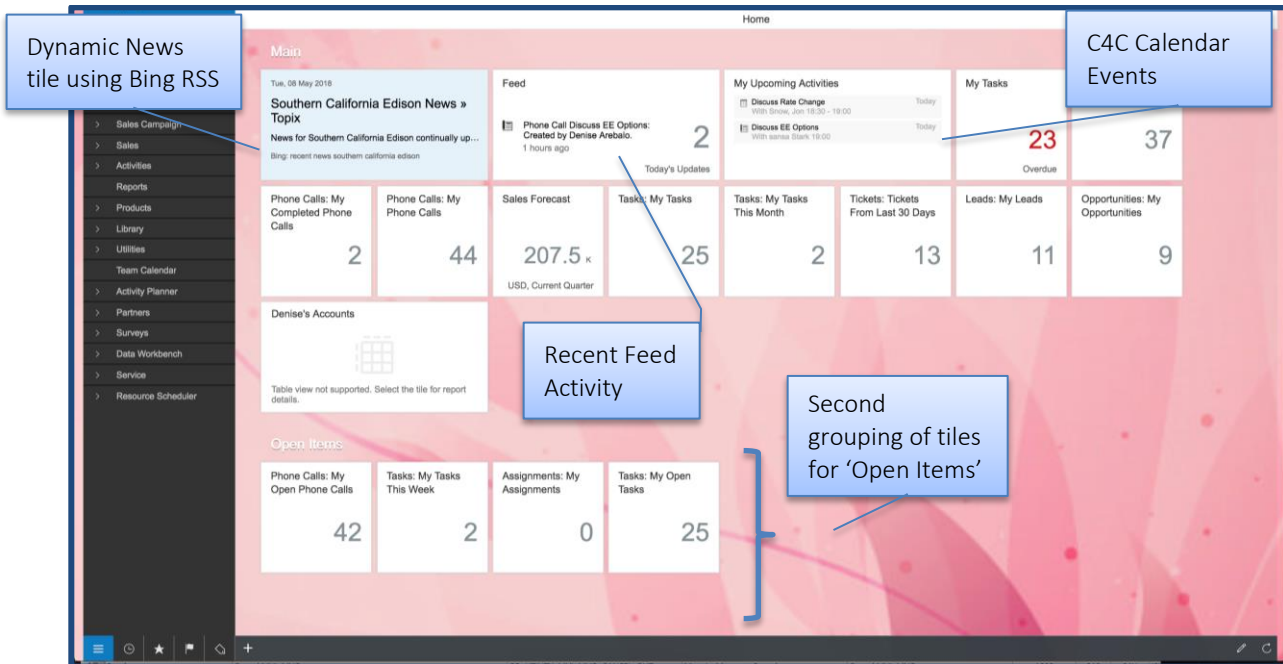


Figure 4: Sample Home Page with personalized tiles and background image

2.1.2.3 UI and General Navigation - Feed

The C4C feed provides users the ability to view updates on items that they are following (to be discussed in the next section). The feed mimics the functionality of common social media platforms by displaying updates in a sequence that users can quickly review throughout the day as shown in **Figure 5**. Users have the ability to post updates within their feed and tag other employees, or C4C objects within the post. Users have the ability to filter and personalize their feed sources so that only certain events are displayed on their feed. In addition, users can personalize their feed so that they can receive an automated system emails for specific updates (Leads, Opportunities, etc.), as shown in **Figure 6**, which provides them a list of the activity from their feed.

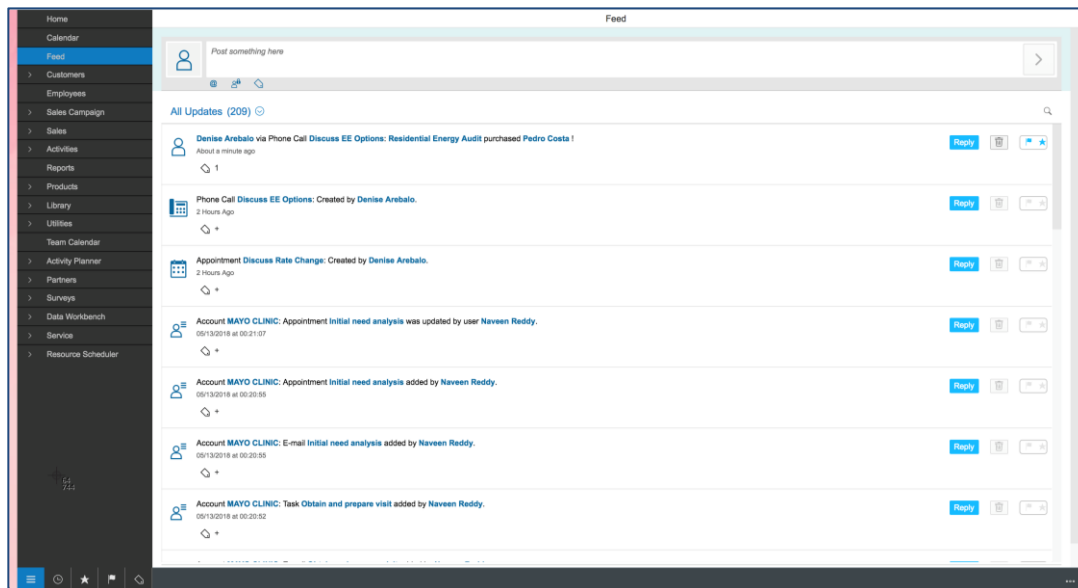


Figure 5: Sample Feed

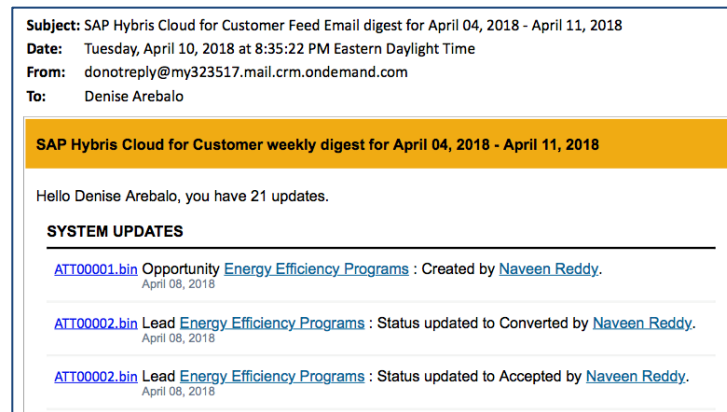


Figure 6: Sample Feed Digest Email

2.1.2.4 UI and General Navigation - Follow, Favorites, Flag, Hashtag and Posting an Update

Hybris C4C provides users the functionality to group and monitor specific accounts, customers, or C4C objects for updates by providing multiple options, described below.

Function	Description	C4C Icon
Follow	Users have the ability to follow accounts or C4C objects (ex: C4C ticket, activity, lead, or account). This functionality can be used by users to follow assigned accounts, or a call center manager to follow a ticket until it has been resolved.	
Favorites	Designating an account or C4C object as a 'Favorite' and by 'Flagging' it, has the same result; it adds the object to the user's personal pane so that it can be quickly accessed. Therefore, users can leverage both functionality, but differentiate their list by type of object. For example, designate only customers or accounts as favorites, and use the Flag to follow tickets, activities, or sales leads. Users can view their favorites by accessing the pane on the bottom left, as shown in Figure 7 .	
Flag	Designating an account or C4C object as a 'Favorite' and by 'Flagging' it, has the same result; it adds the object to the user's personal pane so that it can be quickly accessed. Therefore, users can leverage both functionality, but differentiate their list by type of object. For example, designate only customers or accounts as favorites, and use the Flag to follow tickets, activities, or sales leads. Users can view their flagged items by accessing the pane on the bottom left, as shown in Figure 7 .	
Hashtag	Users are able to add tags to accounts, customers, or C4C objects. This allows users to group multiple activities together for the ability to easily search when needed. Users can search for the hashtags by accessing the pane on the bottom left as shown in Figure 7 .	
Post an Update	Users can post updates from within a C4C object that will appear within the feed. Users can tag other employees or another object within the update. (See Figure 8)	

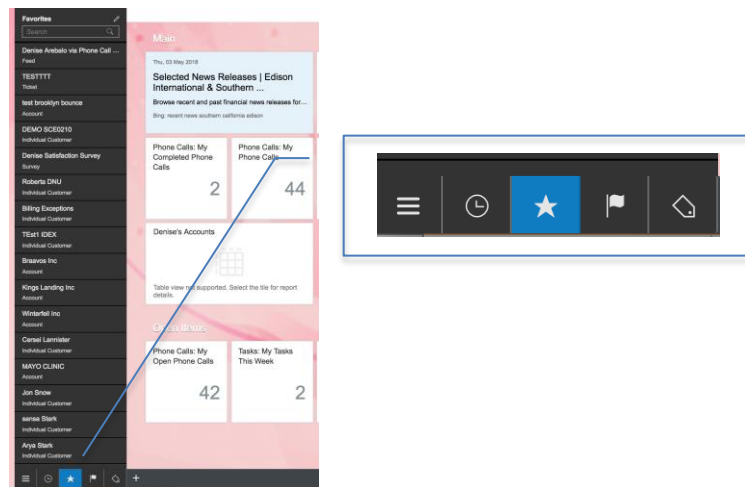


Figure 7: Use the Panes on the bottom left to switch between favorites, flagged items, and hashtags

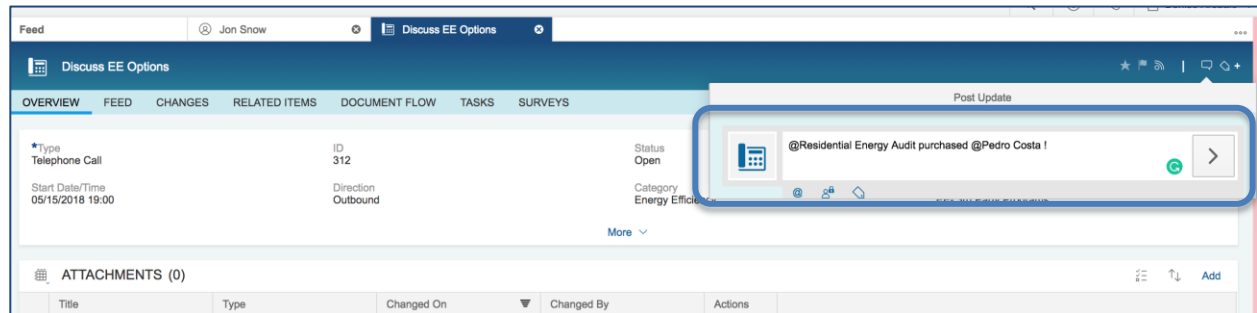


Figure 8: Posting an Update from within a phone call activity

2.1.2.5 UI and General Navigation - Notifications

C4C delivers a standard notifications area that alerts users when they have a C4C object that gets routed to them (assigned), or becomes past due, as shown in **Figure 9**. These notifications can be configured using the 'Workflow Rules' facet within the HTML5 UI. Alerts can be configured to be generated based on different events within C4C, but can only be generated for C4C objects.

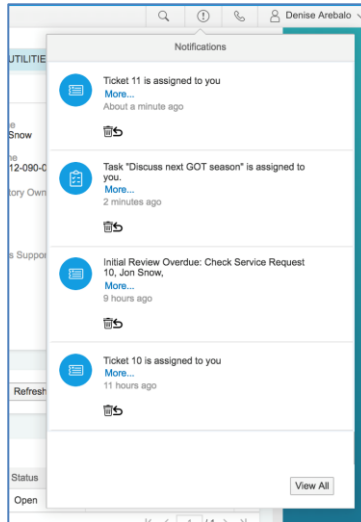


Figure 9: User Notifications

2.1.2.6 UI and General Navigation - Personalization

In addition to the home page personalization, (discussed in Section 2.1.2.2) end users will have the ability to modify the layout of their RUI by hiding/displaying fields that are available, and rearranging their facets (Tabs) as shown in **Figure 10**.

Note: User personalization is only allowed for data the user is authorized to view. If a facet (Tab) is hidden from a user due to authorization controls, they will not be able to expose the facet (tab) using personalization.

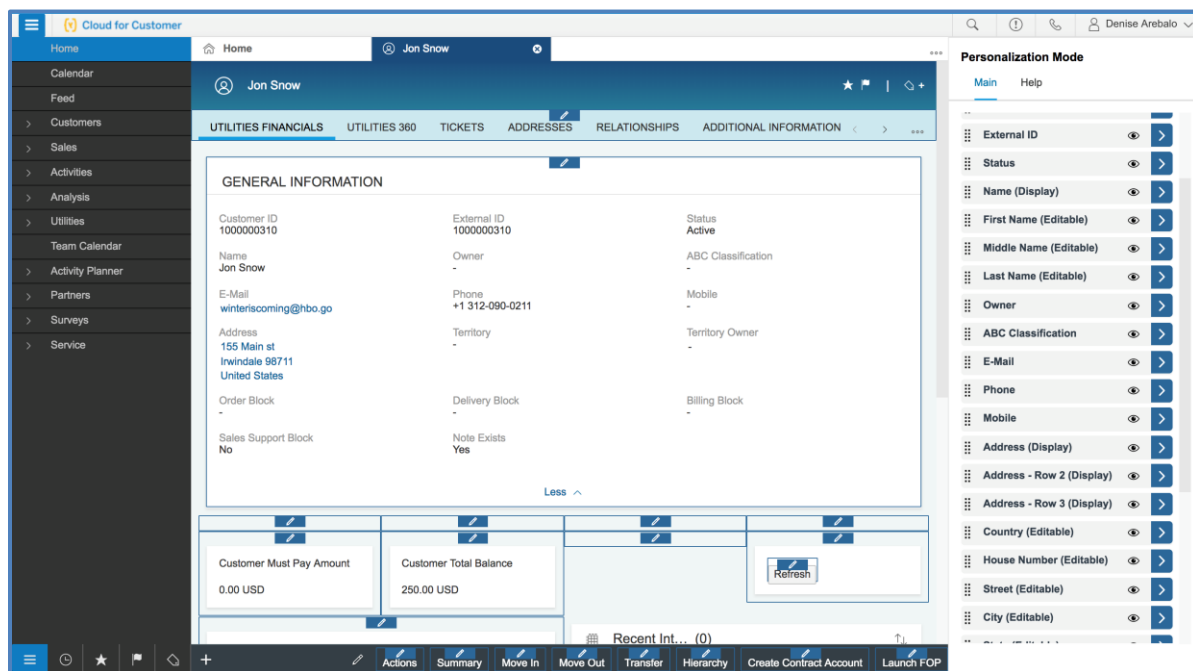


Figure 10: Personalization within the RUI

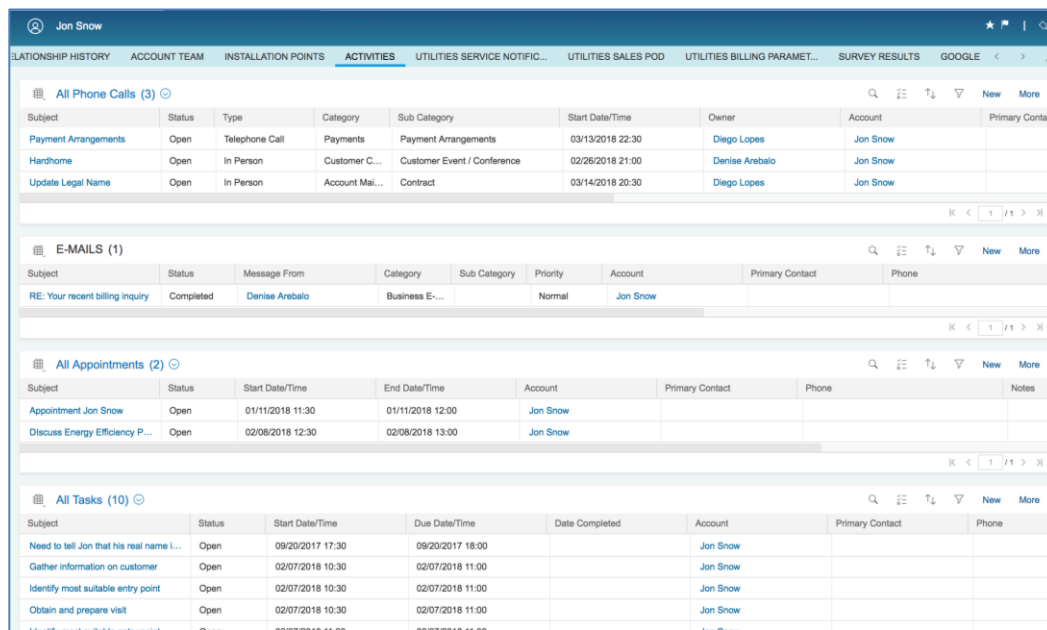
2.1.3 Hybris Cloud for Service

The Hybris Cloud for Service module will be used for Activity Management, Tickets Management ([CIM-CS-08 - Complaints, Claims, and Survey](#)), and Utilities Call Center ([CIM-CS-13 - Manage Inbound Telephone Call - IVR](#)). This document will only provide an overview of this functionality, detailed functionality is covered in multiple BPD's which are referenced.

2.1.3.1 Hybris Cloud for Service - Activity Management

Activities will be used by SCE to record customer interactions. The type of activities created depends on the type of interactions, which are shown in **Figure 11**. A customer's activities can be viewed through the Activities facet (Tab). For additional information on this process, refer to [CIM-CS-12 - Manage Customer Interactions](#).

- **Phone Calls:** Activity will be created to record after call wrap up and/or input notes once a call with the customer has been completed.
- **Emails:** Activity will be created when a user generates an email to a customer through C4C. While this functionality is available, at the time of writing, SCE is planning on using the Opentext Empower Editor for Call Center Agent communications, and BCD is exploring the Outlook /C4C plug-in for sending outbound communications, refer to [CIM-CS-16 Manage Leads, Opportunities, and Territory](#) and [CIM-CS-11 - Manage Outbound Request](#)
- **Appointments:** An appointment is a planned meeting with a customer. Appointments are created prior to a meeting with the customer, and are visible within the C4C calendar. This will be used by the BCD group for their assigned/unassigned customers. For additional information on activities, please refer to [CIM-CS-16 Manage Leads, Opportunities, and Territory](#)
- **Tasks:** A task can be created for oneself, or for another user with the intent of tracking the activity to completion. An example would be if a customer request additional information or action item is noted during an in-person visit, creation of the tasks allows for the action to be tracked to completion. For additional information on activities, please refer to [CIM-CS-16 Manage Leads, Opportunities, and Territory](#)



Jon Snow									
RELATIONSHIP HISTORY ACCOUNT TEAM INSTALLATION POINTS ACTIVITIES UTILITIES SERVICE NOTIFIC... UTILITIES SALES POO UTILITIES BILLING PARAMET... SURVEY RESULTS GOOGLE < >									
All Phone Calls (3)									
Subject	Status	Type	Category	Sub Category	Start Date/Time	Owner	Account	Primary Contact	
Payment Arrangements	Open	Telephone Call	Payments	Payment Arrangements	03/13/2018 22:30	Diego Lopes	Jon Snow		
Hardhome	Open	In Person	Customer C...	Customer Event / Conference	02/26/2018 21:00	Denise Arebalo	Jon Snow		
Update Legal Name	Open	In Person	Account Mai...	Contract	03/14/2018 20:30	Diego Lopes	Jon Snow		
E-MAILS (1)									
Subject	Status	Message From	Category	Sub Category	Priority	Account	Primary Contact	Phone	
RE: Your recent billing inquiry	Completed	Denise Arebalo	Business E...		Normal	Jon Snow			
All Appointments (2)									
Subject	Status	Start Date/Time	End Date/Time	Account	Primary Contact	Phone	Notes		
Appointment Jon Snow	Open	01/11/2018 11:30	01/11/2018 12:00	Jon Snow					
Discuss Energy Efficiency P...	Open	02/08/2018 12:30	02/08/2018 13:00	Jon Snow					
All Tasks (10)									
Subject	Status	Start Date/Time	Due Date/Time	Date Completed	Account	Primary Contact	Phone		
Need to tell Jon that his real name I...	Open	09/20/2017 17:30	09/20/2017 18:00		Jon Snow				
Gather information on customer	Open	02/07/2018 10:30	02/07/2018 11:00		Jon Snow				
Identify most suitable entry point	Open	02/07/2018 10:30	02/07/2018 11:00		Jon Snow				
Obtain and prepare visit	Open	02/07/2018 10:30	02/07/2018 11:00		Jon Snow				

Figure 11: Activities Facet (Tab)

2.1.3.2 Hybris Cloud for Service – Ticket Management

Tickets are used within C4C to record and track work that needs to be performed. At SCE, tickets will be used to track work among agents. If routing to the back-office is required, the users will create a ticket in C4C, as shown in **Figure 12**. Then, the user will generate the BPEM case from within the service ticket by selecting the case category, saving, then selecting 'Generate Exception' from the *Actions* button as shown in **Figure 13**. The enhancement E-CIM-CS-17 BPEM Case Creation has been identified to enable BPEM case creation from C4C. A user is only able to generate one BPEM case per ticket. For additional information on this process, please refer to CIM-CS-06 - Handle Customer Inquiries and [CIM-CS-07 - Handle Bill Inquiries](#)

The screenshot shows a 'Create Service ticket' form. The form is divided into several sections. On the left, there are fields for ID (487), Source (Manual data entry), Object Category, Team (HCL Service Org), Subject (Customer needs to have their bill reviewed), and Description (Please add a note). On the right, there are fields for Priority (Normal), Ticket Category (Billing), Resolution Category (Verify Move in/out Date), Assign To (Pedro Costa), and Requested Start Date (05/17/2018 00:00:00).

Figure 12: Create Service ticket

The screenshot shows the Hybris Cloud for Service interface. The 'OVERVIEW' tab is active, displaying various fields for a service ticket. The 'UTILITIES - EXCEPTION MANAGEMENT' section shows 'Case Category' as 'Business Partner BPEM'. The 'CATEGORY' section shows 'Service Category' as 'Credit'. The 'DESCRIPTION' section is empty. The 'ITEMS' tab is also visible, showing a timeline of events. An orange arrow points from the 'Generate Exception' button in the 'UTILITIES - EXCEPTION MANAGEMENT' section to the 'Generate Exception' button in the 'ITEMS' section.

Figure 13: Generate BPEM Case within a Ticket

2.1.4 Utilities Call Center

The functionality within the Utilities Call Center includes four utilities specific facets (tabs), which are detailed in this section.

2.1.4.1 Utilities Call Center- Utilities Financials

The Utilities Financials facet (tab) will be the landing page when a customer's account is accessed by the users. This facet (tab) shows a detailed overview of the SCE business partner; invoices, payments, open items, installment plans, and payment plans, in addition to the premise and contracts associated with the business partner, as shown in **Figure 14**. Users can update meter readings, view invoices, or enroll the customer in a budget billing plan or installment plan.

Contract Account External ID	Document No	Amount	Net Due Date	Days Past Due Date	Locked Status	Dunning Level	Dunning %
1000000001	810100000001	200.00 USD	01/01/2017	459			
1000000001	840100000002	200.00 USD	06/01/2018				
1000000001	860100000003	125.41 USD	04/11/2018	34			

Figure 14: Utilities Financials

The Utilities Financials facet (tab) has the following sections that are displayed:

- **Open Items:** This section displays unpaid items that are grouped and displayed for each contract account. If needed, the user can set an individual item lock to prevent dunning or clearing.
- **Premises:** Displays the list of premises where the customer has an active contract.
- **Installment Plans:** Installment Plans can be issued to a customer via SCE.com, IVR or by calling and requesting through an authorized Front Office or Back Office agent. Installment plans can be displayed, modified and deactivated from within C4C, as shown in **Figure 15**. When an inbound call is received to request a payment arrangement, the front office agent will review the business partner's overall credit profile to validate that the customer qualifies for an installment plan based on SCE's business rules and credit policy. If the customer is deemed eligible for payment arrangements, the front office agent can select the appropriate Installment plan based on SCE Business Rules. Please reference [BPD-CC-FI-08 - Manage Collections \(Dunning\) - Manage Payment Arrangements Status](#) for additional information.

Invoice Payments Open Items Installment Plan Payment Plan Premise Contracts with Meter Reading						Edit Installment Plan
(6)						Deactivate Installment Plan
Installment Plan No	Contract Account External ID	Total	No of Installments	Status	Start Date	
66000032	25063371	0.00 US Dollar	2	Open	01/23/2018	
66000029	25063371	40.83 US Dollar	3	Cancelled	08/15/2017	
66000028	25063371	40.83 US Dollar	4	Cancelled	07/05/2017	
66000027	25063371	40.83 US Dollar	3	Cancelled	05/11/2017	

Figure 15: Installment Plan

- **Payments:** A list of the customer's recent payments including the posting date and cleared amount are displayed (see **Figure 16**).

Invoice	Payments	Open Items	Installment Plan	Payment Plan	Premise	Contracts with Meter Reading
Payments (1)						
Posting Date	Payment Document No	Amount Paid	Payment Origin	Contract Account External ID	Cleared Amount	
05/13/2018	540100000026	25.00 USD	Manual Payment	200000000149	-25.00 USD	

Figure 16: Payments

- **Payment Plan:** A Payment Plan is a budget billing plan (also known as level pay plan) that offers eligible customers the opportunity to pay for their annual service across 11 equal monthly payments. The first month of the plan is called “Month 1”. “Month 12” is the month that the customer receives the settlement bill. Payment Plans are visible and can be created in C4C as shown in Figure 17. For additional information regarding Budget Billing, please refer to [BPD UB-BI-11 - Manage Bill Arrangements](#).

Invoice

Payments

Open Items

Installment Plan

Payment Plan

Premise

Contracts with Meter Reading

Payment Plan (1)

Create Payment Plan

Status	Contract Account Ext. ID	Contract ID	Division	Curr. diff. amt	Move-In Date	Payment Plan No.	Start Date
	25063371	3011389	Electric	0.00 US Dollar	01/01/2015	60000028	02/01/2018

Figure 17: Payment Plans

- **Contracts with Meter Reading:** This section displays all contracts for the customer with their monthly meter readings for Mechanical Meters. The user can select a row to display all recorded meter readings for the selected contract account. This is only applicable for non-AMI meters. At the time of writing, AMI functionality is not yet available in C4C. However, this functionality is expected to be delivered in phases over the next releases (1808, 1811, and 1902).

Invoice

Payments

Open Items

Installment Plan

Payment Plan

Premise

Contracts with Meter Reading

Contracts (1)

🔍 ↕ ⚙

Contract Account External ID	Contract ID	Service Address	Installation ID	Division	Move-In Date	Move-Out Date	Billing Block	Release
700000000016	3000000187	150 Main st Irwindale CA U...	7000000349	Electric	01/01/2017	Unlimited		

1 selected | [Clear Selection](#)

⏪ < 1 / 1 > ⏩

Meter Readings (0)

↕ ⚙

[New Meter Reading](#)

[More](#)

MR Order	Division	Device ID	Meter Active	Meter Reading T	Date	Meter Reading	Unit of Measure	Meter Reading F	Note	Meter Reading E	Expected
----------	----------	-----------	--------------	-----------------	------	---------------	-----------------	-----------------	------	-----------------	----------

No data

Figure 18: Contracts with Meter Reading

2.1.4.2 Utilities Call Center- Utilities 360

The Utilities 360 facet (Tab) has three core components (refer to **Figure 19**):

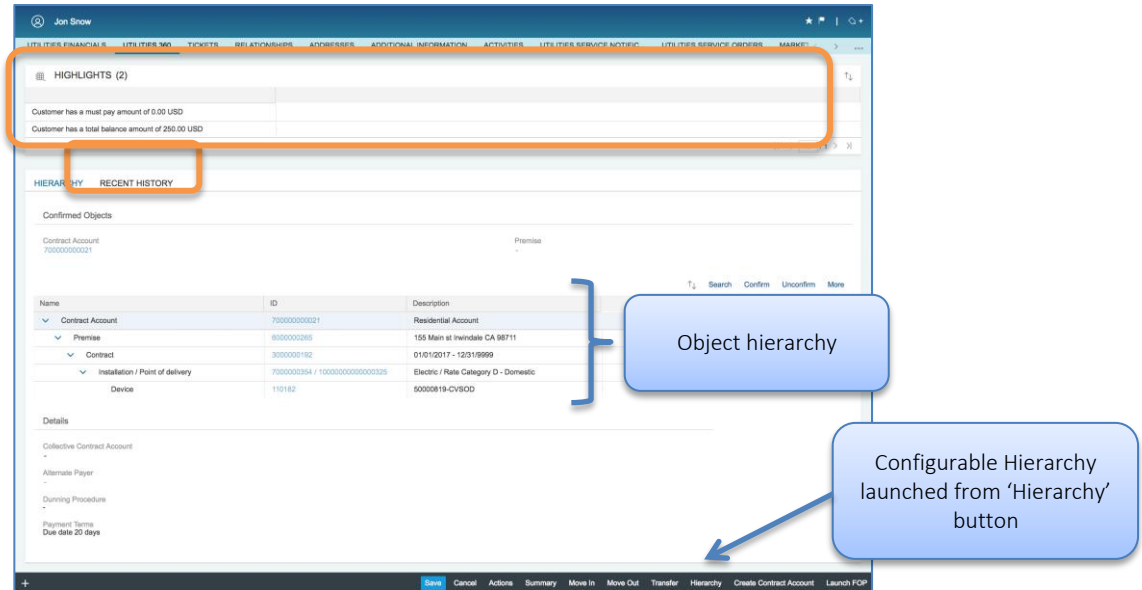


Figure 19: Utilities 360

- **Highlights** - The highlights section is at the top, and by default, will display the customer's total balance and overdue balance.
- **Hierarchy** - The Utilities 360 facet (Tab) provides the user with the object hierarchy including both business master data and a few of the key S4/HANA technical master data objects: the premise, installation, and device. The user can click on each row within the hierarchy to view additional information on the selected object directly below the hierarchy. In addition, the user will be able to view the parent/child relationships between collective accounts, as shown in **Figure 20a**. While the embedded hierarchy within the Utilities 360 is not configurable, an additional hierarchy is available, which will be configured for SCE to display specific attributes, such as fringe account,. The configurable hierarchy (as shown in **Figure 20b**) can be accessed via the 'Hierarchy' button near the bottom right of the screen.

Name	ID	Description	
Collective Contract Account	300000000000	Collective Invoice Test	
Contract Account	200000000092	Child Account 01	
Premise	6000000089	110 Main st Irwindale CA 98711	
Contract	3000000034	01/13/2018 - 12/31/9999	
Installation / Point of delivery	7000000107 / 1000000000000033	Electric / Rate Category D	
Device	110037	50000819-CVSOD	
Contract Account	200000000104	Child Account 03	
Premise	6000000091	110 Main st Irwindale CA 98711	
Contract	3000000046	01/31/2018 - 12/31/9999	
Installation / Point of delivery	7000000109 / 1000000000000037	Electric / Rate Category D	
Device	110039	50000819-CVSOD	

Figure 20a: Embedded Hierarchy with Collective Account

Configurable Object Hierarchy					✕	
Contract Account:						
CONFIGURABLE HIERARCHY (1)					Expand All	Collapse All
	Type	ID	Description			
▼		PARTNER	1000000305	TestData SecurityDeposit01 / 150 Main st / Irwindale 98711		
▼		ACCOUNT	700000000016	01: TestData SecurityDeposit01/Residential Account		
▼		PREMISE	6000000260 FRINGE ACCOUNT	Main st 150 Apt 3, Irwindale		
▼		CONTRACT	3000000187 SCE_DIST	01/01/2017 12/31/9999		
▼		INSTALLATION	7000000349 ELECTRIC	CTLA		
▼		DEVICE	110177	240 10S 20 3.6 27-7/9 MAG KWH		

Figure 20b: Configurable Hierarchy

Recent History - This provides an overview of the most recent tickets, activities, invoices, payments, business partner contacts, and service orders created in one centralized view (see **Figure 21**).

Feed

Jon Snow

UTILITIES 360TICKETSUTILITIES CONTRACTSUTILITIES SERVICE ORDERSUTILITIES FINANCIALSADDRESSESRELATIONSHIPS

SALES DA < > >>>

HIERARCHY

RECENT HISTORY

Tickets (13)

1

3

>

>>>

Ticket ID	Description	Priority	Status	Created On	Assigned To	
214	testing	Normal	Open	02/06/2018 10:32 ...	Denise Arebalo	
224	test	Normal	Open	02/07/2018 01:14 ...	Denise Arebalo	
226	test	Normal	Open	02/08/2018 10:47 ...	Denise Arebalo	
227	tester	Normal	Open	02/08/2018 10:49 ...	Denise Arebalo	
228	test	Normal	Open	02/09/2018 14:27 ...	Denise Arebalo	

Activities (18)

1

5

>

>>>

ID	Activity Type	Subject	Start Date Time	End Date Time	Organizer/Processor	Phone
63	Appointment	Discuss Energy Efficiency P...	02/08/2018 12:30	02/08/2018 13:00	Denise Arebalo	
127	Phone Call	Payment Arrangements	03/13/2018 22:30	03/13/2018 23:00		
110	Phone Call	Hardhome	02/26/2018 21:00	02/26/2018 21:30		
31	Activity Task	Need to tell Jon that his real ...	05/15/2018 21:30	05/15/2018 22:00		

Invoices (0)

1

5

>

>>>

Invoice Number	Contract Account External ID	Posting Date	Amount Due	Due Date	
----------------	------------------------------	--------------	------------	----------	--

No data

Payments (0)

1

5

>

>>>

Contract Account External ID	Payment Document No	Description	Confirmation	Posting Date	Amount
------------------------------	---------------------	-------------	--------------	--------------	--------

No data

Figure 21: Recent History

2.1.4.3 Utilities Call Center- Utilities Service Orders

Front office agents will often need to create a field requests on behalf of a customer or respond to customer inquiries regarding the status of an outstanding field request. The Utilities Service Orders facet (Tab) in **Figure 22**, will be used to view the status of an order, to issue a new order, and to modify or cancel existing orders. This view shows only orders relevant to confirmed customer. To modify or cancel an order, select the order and click the 'More' button. To create service orders from C4C, the enhancement E-CIM-CS-17 Service Order Creation has been identified.

UTILITIES FINANCIALS UTILITIES 360 TICKETS RELATIONSHIPS ADDRESSES ADDITIONAL INFORMATION ACTIVITIES UTILITIES SERVICE NOTIFIC... UTILITIES SERVICE ORDERS MARKET < > ...				
Service Orders (8)				
Service Order ID: 5000011 Status: Initial	Description: TUON-TURN ON Premise External ID: --	Service Order Type: TUON Premise Address: 3 Main st Irwindale US 98711	Created On: 04/12/2018 Equipment: --	Edit Cancel
Service Order ID: 5000009 Status: Initial	Description: TUON-TURN ON PEDRO TEST SERVICE ORDER TU Premise External ID: --	Service Order Type: TUON Premise Address: 3 Main st Irwindale US 98711	Created On: 04/10/2018 Equipment: --	
Service Order ID: 5000008 Status: Initial	Description: TUON-TURN ON Premise External ID: --	Service Order Type: TUON Premise Address: 3 Main st Irwindale US 98711	Created On: 04/10/2018 Equipment: --	
Service Order ID: 5000007 Status: Initial	Description: TUON-TURN ON Premise External ID: --	Service Order Type: TUON Premise Address: 3 Main st Irwindale US 98711	Created On: 04/09/2018 Equipment: --	
Service Order ID: 5000006 Status: Cancelled	Description: TUON-TURN ON Premise External ID: --	Service Order Type: TUON Premise Address: 3 Main st Irwindale US 98711	Created On: 04/09/2018 Equipment: --	
Service Order ID: 5000005 Status: Initial	Description: PTLTNOLT-PART LIGHT/NO LIGHT Premise External ID: --	Service Order Type: TRBL Premise Address: 3 Main st Irwindale US 98711	Created On: 04/04/2018 Equipment: --	
Service Order ID: 5000004 Status: Initial	Description: PTLTNOLT-PART LIGHT/NO LIGHT Premise External ID: --	Service Order Type: TRBL Premise Address: 3 Main st Irwindale US 98711	Created On: 04/03/2018 Equipment: --	
Service Order ID: 5000002 Status: Dispatched	Description: TUON-TURN ON Premise External ID: --	Service Order Type: TUON Premise Address: 3 Main st Irwindale US 98711	Created On: 03/28/2018 Equipment: --	
1 selected Clear Selection				

Figure 22: Utilities Service Order Facet

If the 'Edit' option is selected, a pop-up will be generated to modify the request date (Figure 23). If the 'Cancel' option is selected, a pop-up to confirm the cancellation will appear (Figure 24). For additional information on the service order process, please refer to [BPD CIM-CS-04 Handle Trouble & Outage Contacts - Process Trouble Call](#) and [BPD CIM-CS-05-Handle Request-Process Field Work Request](#).

3 Main st Irwindale US 98711

Edit Service Order

Service Order ID
5000011

Service Address
3 Main st Irwindale US 98711

Status
Initial

Start Date
05/04/2018 5:38 AM

End Date
05/04/2018 5:38 AM

Description
TUON-TURN ON

Save Cancel

Premise External ID:
--

Premise Address:
3 Main st Irwindale US 98711

Equipment:
--

Figure 23: Edit Service Order

TUON-TURN ON PEDRO TEST SERVICE ORDER TU TUON

Cancel Service Order

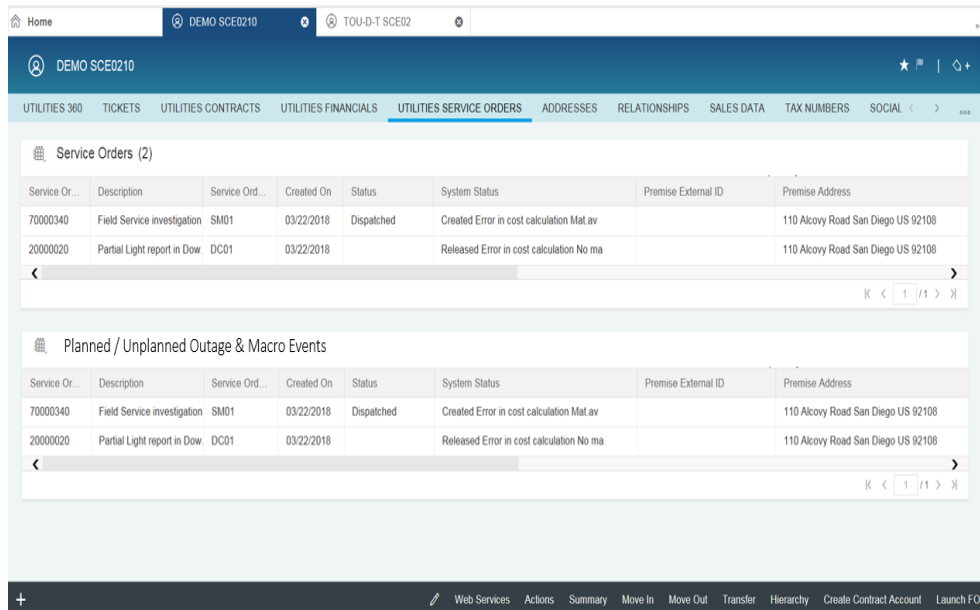
Do you really want to cancel the service order?

OK Cancel

3 Main st Irwindale US 98711

Figure 24: Cancel Service Order

In order to view planned, unplanned and macro outage events, an enhancement E-CIM-CS-17 C4C Outage View has been identified which will enhance the Utilities Service Orders facet (tab), as shown in **Figure 25**. A front office process has been identified which will allow users to query S4/HANA to view open service orders.



The screenshot displays the 'Utilities Service Orders' tab in the E-CIM-CS-17 C4C Outage View. It features two main sections: 'Service Orders (2)' and 'Planned / Unplanned Outage & Macro Events'. Both sections contain a table with columns: Service Or..., Description, Service Ord..., Created On, Status, System Status, Premise External ID, and Premise Address.

Service Or...	Description	Service Ord...	Created On	Status	System Status	Premise External ID	Premise Address
70000340	Field Service investigation	SM01	03/22/2018	Dispatched	Created Error in cost calculation Mat av		110 Alcovy Road San Diego US 92108
20000020	Partial Light report in Dow	DC01	03/22/2018		Released Error in cost calculation No ma		110 Alcovy Road San Diego US 92108

Below the tables, there is a navigation bar with options: Web Services, Actions, Summary, Move In, Move Out, Transfer, Hierarchy, Create Contract Account, and Launch FOP.

Figure 25: Macro Outage Event Enhancement

2.1.4.4 Utilities Call Center - Move-In / Move-Out / Transfer

The move-in/out/transfer process will be performed via C4C. The outline of each process is below. For additional details, please refer to [Process - 15 - CIM-CS-02 - Handle Customer Move-in/Out - Contract New](#) and [Process - 16 - CIM-CS-03 - Handle Customer Move-in/Out - Contract End](#).

Move-In: Once a user has confirmed the customer, the user can perform the guided move-in procedure. The move-in will be initiated via the “Move-In” button. The move-in consists of 4 steps:

1. **Customer & Premise:** This allows the user to select the desired premise and move-in date.
2. **Service:** The user will select the relevant installations that the customer is selecting to have turned-on/energized.
3. **Contract Account & Billing:** The user will have the ability to either select an existing contract account, or create a new contract account, and specify the billing address.
4. **Review:** The user will review the selections with the customer, and then create the move-in.

Move-Out: Once the user has confirmed the customer, the user can perform the guided move-out procedure. The move-out will be initiated via the “Move-Out” button. The move-out consists of 4 steps:

1. **Customer & Premise:** This allows the user to select the move-out premise and desired date.
2. **Service:** The user will select the relevant services the customer is choosing to terminate.
3. **Contract Account & Billing:** The user will have the ability to update their mailing address for the final bill.
4. **Review:** The user will review the selections with the customer, and then create the move-out.

Transfer: The transfer process is a combination of both a move-out and a move-in performed at the same time. Once you have confirmed your customer, the transfer guided procedure can be initiated by clicking the 'Transfer' button. The transfer process consists of 6 steps:

1. **Move-Out:** In this step, the user will select the customer's desired move-out date and premise.
2. **Move-Out Services:** The user will select the relevant services the customer is choosing to terminate.
3. **Move-In:** In this step, the user will select the customer's desired move-in date and premise.
4. **Move-In Services:** The user will select the relevant installations that the customer is selecting to have turned-on/energized.
5. **Contract Account & Billing:** The user will select an existing contract account or create a new contract account and maintain the mailing address.
6. **Review:** The user will review the selections with the customer, and create the transfer of service.

Credit Check

An external credit check to Experian is used to determine a customer's risk class, which is based on their credit score. For information regarding the risk class to credit score mapping, please refer to BPD [CC-FI-21 – Manage Credit – Credit Score](#). The risk class is then used to assess the amount of the deposit on move-in. For additional information regarding security deposit calculation, please refer to [CC-FI-03 Manage Credit – Manage Security Deposit](#).

The credit check functionality will be available to front office agents through enhancement [1805 - E-CIM-CS-17 - Credit Check](#) and [I-CC-FI-03 Credit Check](#). In order to trigger the credit check, the BP must have the Social Security number populated (refer to BPD [CIM-CS-01 Manage Business Partner](#)).

While the credit check will be done outside of the C4C move-in guided process, the credit check are generally done as a part of a customer requested move-in, or ad-hoc when requested by the customer. Customers will generally request an ad-hoc credit check in cases where they do not want to pay the quoted security deposit amount, and would rather have the credit check done if they think their credit score will help waive the deposit amount.

Credit Check is not applicable for the following BP types:

- New residential BP without SSN updated on BP record
- Non-residential BP
- Existing BP with a good credit worthiness score; consistent receipt of payment on time.
- Sub-Metered Accounts

The initial credit check will result in one of the following:

Credit Check Result	Expected User Action
Invalid SSN	The SSN is found to be invalid. A BPEM case of "Customer Verification Investigation" is created manually to the Credit Team, The agent will proceed to create the move-in but will manually set the contract billing block, and cancel the service order (if applicable)
Deposit	The security deposit should be charged to the customer.
No Deposit	The security deposit should not be charged to the customer.
Authenticate / Authenticate with Deposit	A secondary call to Experian is required. This is discussed below.
Refer to Credit	A BPEM case of "Customer Verification Investigation" is created manually to the Credit Team. The agent will proceed to create the move-in but will manually set the contract billing block and cancel the service order (if applicable).

Credit Bureau Authentication

The Experian Authentication process is only triggered if the results of the initial credit check are “Authenticate” or “Authenticate with Deposit” is returned. There are two results of the Authentication process are detailed below. The Experian Authentication process be done through [1805 - E-CIM-CS-17 - Credit Check](#) and an interface as defined in [SCC 244 CIM-CS-17 Credit Bureau Authenticate](#).

Authentication Result	Expected User Action
Refer to Credit	The customer did not answer the authentication questions correctly. A BPDM case of “Customer Verification Investigation” is created manually to the Credit Team. The agent will proceed to create the move-in, but will manually add a billing block to the contract and cancel the reconnection service order (if applicable).
Validated	The customer answered the questions correctly. The user will assess/not assess deposit based on initial result (Authenticate with Deposit or Authenticate (without deposit)).

2.1.4.5 Utilities Call Center - Utilities Rate Change

The Utilities Rate Change facet (tab) will be used when a customer request a rate change or when a user needs to display installation facts (billing parameters) on a customer’s contract (**Figure 26**).

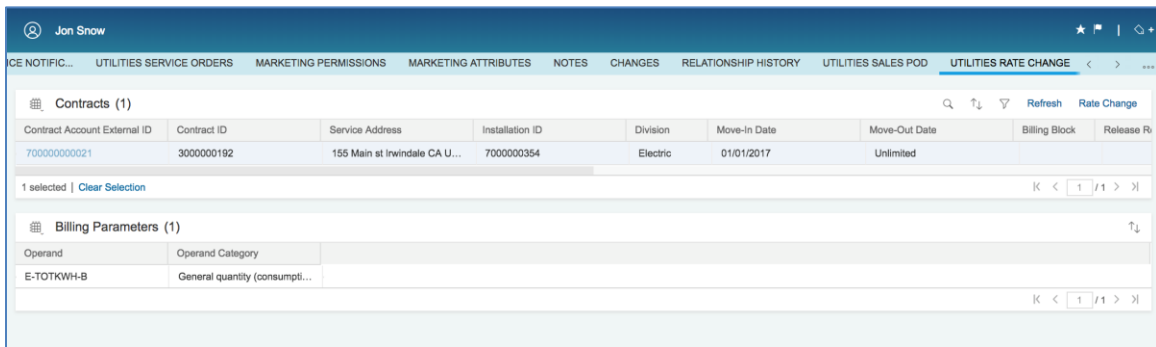


Figure 26: Utilities Rate Change Facet

If the user needs to change the customer’s rate, then they will select the contract and click ‘Rate Change’. A pop-up will be generated which will allow the user to select the new rate (**Figure 27**). When performing a rate change, the validity of the new rate will not be effective until the first day of their next billing period. A FRICEW object has been identified so that the next bill cycle date defaults for the user (E-CIM-CS-02 - Default next bill cycle date).

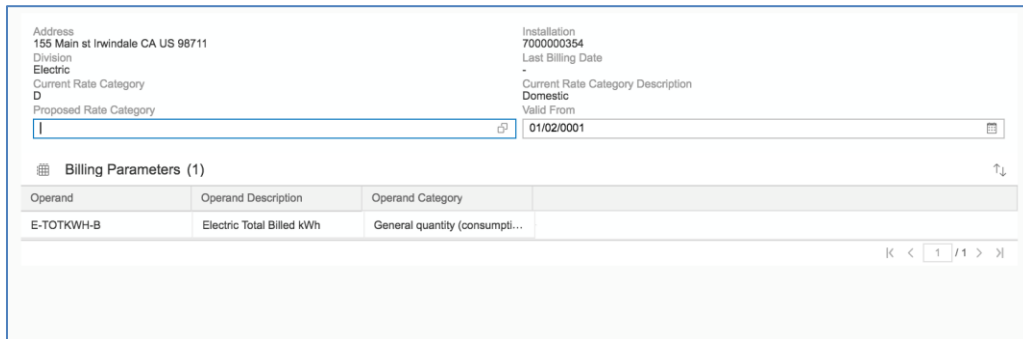
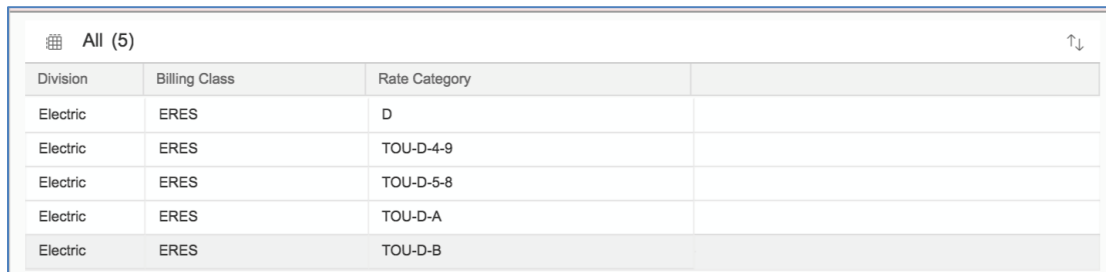


Figure 27: Rate Category Change

Using the field help available in the ‘Proposed Rate Category’ field (**Figure 26**), an additional pop-up will display the available rates. The enhancement E-CIM-CS-17 Rate Change Validations has been identified so that only certain rates are available for

selection per the SCE business rules. Therefore, some rates which require special treatment (like NEM) will be required to be performed via back-office only.

Once the user has selected the new desired rate, they will save the changes using the “Rate Change Using MDT” button.



All (5) ↑↓			
Division	Billing Class	Rate Category	
Electric	ERES	D	
Electric	ERES	TOU-D-4-9	
Electric	ERES	TOU-D-5-8	
Electric	ERES	TOU-D-A	
Electric	ERES	TOU-D-B	

Figure 26: ‘Proposed Rate Category’ Selection Help

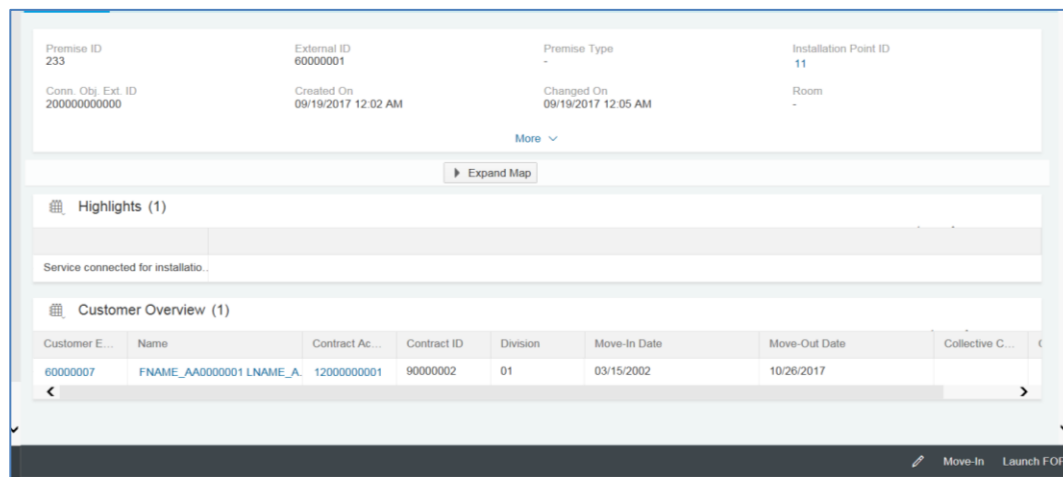
2.1.4.6 Utilities Call Center – Premise

The Premise facet provides users the ability to view details specific to a service location including:

- Overview of the service location address: This will inform the user of the premise type, detailed address, and
- Highlights: This section will inform the user if the location is connected / disconnected
- Customer Overview: This section details the customer that is active at the service location.

In addition, the user can view the service orders that were specifically created for the premise location.

The premise facet can be launched from within a customer’s account (to view the premise details), or a premise can be searched for independently. The move-in process can be initiated from this facet (tab), for additional information please refer to [BPD CIM-CS-02 Handle Customer Move-in Out - Contract New](#).



Premise ID	External ID	Premise Type	Installation Point ID
233	60000001	-	11
Conn. Obj. Ext. ID 200000000000	Created On 09/19/2017 12:02 AM	Changed On 09/19/2017 12:05 AM	Room -

More ▼

▶ Expand Map

Highlights (1)

Service connected for installatio...

Customer Overview (1)

Customer E...	Name	Contract Ac...	Contract ID	Division	Move-In Date	Move-Out Date	Collective C...
60000007	FNAME_AA0000001 LNAME_A...	12000000001	90000002	01	03/15/2002	10/26/2017	

Move-In Launch FOP

Figure 27: Premise Overview

2.1.4.7 Utilities Call Center - Mash-ups and Front Office Processes

C4C offers different mechanisms to allow users to access data from external sources while using the C4C application. Depending on the third party application, the following options are available:

- **Front Office Processes:** A front office process (FOP) can be launched via C4C using the ‘Launch FOP’ button. This is used to launch S4/HANA transactions from C4C. This will allow front office users the ability to access the transaction they need within S4/HANA with minimal navigation as possible. At the time of writing, the full list of is still being compiled and

reviewed. The list provided below is preliminary and not yet finalized. To view the most updated list, please refer to the FOP list within [APSE](#).

S4/HANA Transaction Code	Description
CAA2/CAA3	Display/Change Contract Account: This will be used to view custom fields that are defined on the contract account (i.e for bankruptcy). Please refer to process CC-FI-01 - Business Master Data - Manage Contract Account
FP08	Reverse document: Used to reverse line items, i.e. late payment charges. Please refer to CC-FI-15 - Handle Account Adjustments - Manage Miscellaneous Postings
BP	Change Business partner: This will be used to add bank details on the business partner. Please refer to process CIM-CS-01 Manage Business Partner Data
FP03H	Collection agency assignment history: This will be used to view the balance that has been sent to collection agency. Please refer to process CC-FI-07 - Manage Collections (Dunning) - Manage Collection Agencies .
FPL9	Account Display: This will be used to view miscellaneous charges, return fees, late payment charges. Please refer to process CC-FI-02 - Manage Credit - Manage Account Balance Display
FP04H	Write Off History: This allows the user to view the amount of the balance that was written off, the reason, and the date. Please refer to process CC-FI-06 - Manage Collections (Dunning) - Manage Write Offs .
EK27	Display loan: This will be used to view the customer's OBF loan schedule, including number of installments paid, due dates, and amounts. Please refer to process PP-CS-02_02 - Program Enrollment and Delivery, De-Enrollment & Decommission Programs
FP40	Transfer post: This will be used to transfer balances from one account to another. Please refer to process CC-FI-12 - Handle Account Adjustments (FICA) - Manage Transfer Postings .
FPCHR	Check register: This will be used to display the status of check refunds that are issued to the customer. Please refer to CC-FI-22 - Handle Account Adjustments (FICA) - Manage Customer Refunds
FPM3	Dunning history: This will be used to view details regarding where the customer is within the dunning process. Please refer to process CC-FI-02 - Manage Credit - Manage Account Balance Display
FP30H	Find payments: This will be used to find missing payments. Please refer to process PY-FI-09 Manage Incoming Payments
FPAY1A	Create payment specification: This will be used to take one-time payments. Please refer to process PY-FI-09 - Receive Payments - Manage Incoming Payments
FPM4	Return history: This is used to view customer returned payment history. This will assist in agents determining a customer's eligibility to offer installment plans. Please refer to process PY-FI-11 - Handle Payment Follow-Up - Manage Returns
ES67	Display Device Location: This will be used to view information regarding the customer's meter location, including classification characteristics, ex: site access code, alternate meter location, and meter action. Please refer to process DF-DM-01 - Create/Maintain Technical Master Data - Manage Technical Master Data .

- **Mash-up (URL, web service, or HTML):** A mash-up allows the user to easily launch a third party application via a link or by embedding it within C4C (depends on the mash-up type and third party application). At SCE, the following mash-ups have been identified. The list provided below is preliminary and not yet finalized.

Application	Description
SCE.COM	SPOC
iEnergy	Rebate application program
GridX	Rate comparison tool
DM	Design Manager

2.1.5 Hybris Cloud for Sales

The Hybris Cloud for Sales module is used for sales planning, leads & opportunity management. Users with authorization will have the ability to use both Hybris Cloud for Sales and Service functionality. This document will only provide an overview of this functionality, as it will be covered in detail in BPD [CIM-CS-16 - Manage Leads, Opportunities & Territory](#).

2.1.5.1 Hybris Cloud for Sales- Leads

Leads are the initial step within the sales process to record interest in a specific program or product. Leads can be created manually (**Figure 29**), or created via excel upload as a result of an event. Leads can be qualified directly within C4C then directly converted to an opportunity. Sales phases with mandatory activities that must be completed can be defined. This will assist in standardizing the sales process to ensure each team member is performing the required tasks. For additional information regarding the leads management process, please refer to [CIM-CS-16 - Manage Leads, Opportunities & Territory](#).

Figure 29: Leads Creation

2.1.5.2 Hybris Cloud for Sales - Opportunities

Once a lead is qualified, it can be converted to an opportunity. Information from the lead will be copied to the opportunity, and the user will be able to set multiple attributes including the forecast category, source of the opportunity, probability, expected value, and whether or not it should be included in the forecast (**Figure 30**). For additional information regarding the opportunity management process, refer to [CIM-CS-16 - Manage Leads, Opportunities & Territory](#).

Document Type	*Name
Opportunity	New Opportunity
*Account	Primary Contact
Braavos Inc	
Source	Expected Value
Campaign	100,000.00 USD
Start Date	Close Date
05/15/2018	11/11/2018
Sales Cycle	Sales Phase
General opportunity	Identify opportunity
Probability	Publish to Forecast
25%	NO
Forecast Category	Category
Pipeline	
Campaign	*Owner
	Edmund Wan
Note	

Figure 30: Creation of an Opportunity

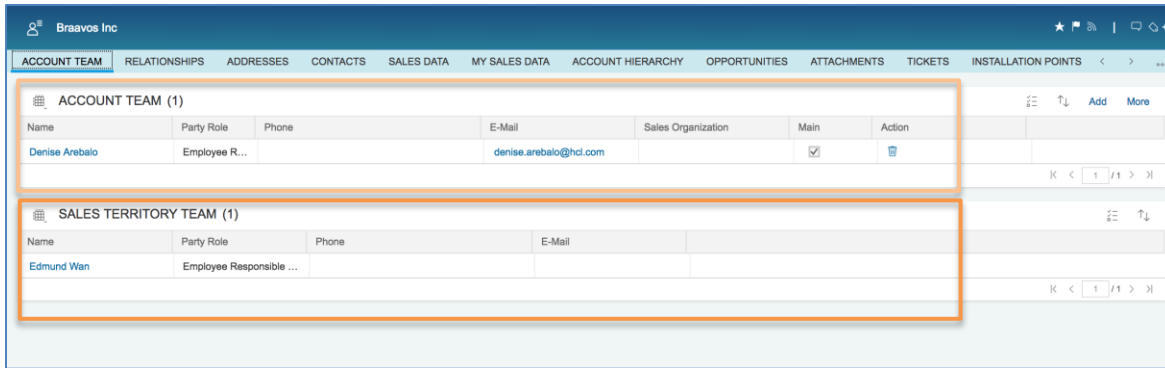
2.1.5.3 Hybris Cloud for Sales - Territory Management

Territory management is used as a mechanism to allocate a group of accounts to dedicate a resource within a sale organization or account team. Realignment runs are periodically executed to reassign accounts based on changes in both the customer account data, and potential changes in the assigned account teams. If an account is assigned it will be denoted in the customer's account details, including their territory assignment (**Figure 31**).

External ID	Account ID	Name	Status
1008351	1008351	Braavos Inc	Active
Prospect	*Role	Parent Account	Address
No	Customer	-	Sanford NC 27332
Phone	Fax	Web Site	United States
-	-	-	Main Contact
Owner	Override Territory	Territory	Territory Owner
Denise Arebalo	No	Value Accounts	Edmund Wan
ABC Classification	Nielsen ID	Industry	YTD Revenue
C-Value	-	-	0.00 USD
Active Pipeline	D-U-N-S	Contact Permission	Billing Block
0.00 USD	-	-	-
Note	Segment Identifier		
	-		

Figure 31: Account Team and Territory Assignment

The Account Team is for employees that are assigned to an account, regardless of modifications in the territory rules and realignment. The account 'Owner', shown below, is the employee responsible, but it is not assigned through territory management. The "Sales Territory Team" is the team that is assigned as a result of territory management, as shown in **Figure 32**. For additional information on territory management, please refer to [CIM-CS-16 - Manage Leads, Opportunities & Territory](#).



The screenshot shows the Braavos Inc. interface with two tables highlighted by orange boxes. The first table, 'ACCOUNT TEAM (1)', lists Denise Arebalo as an Employee R... with email denise.arebalo@hcl.com. The second table, 'SALES TERRITORY TEAM (1)', lists Edmund Wan as an Employee Responsible ...

Name	Party Role	Phone	E-Mail	Sales Organization	Main	Action
Denise Arebalo	Employee R...		denise.arebalo@hcl.com		☒	

Name	Party Role	Phone	E-Mail
Edmund Wan	Employee Responsible ...		

Figure 32: Account Team

2.1.6 Organizational Structure and Roles

This section will discuss the business roles and organizational structure that will be configured in C4C.

2.1.6.1 Organizational Structure and Roles - Organizational Structure

SCE currently performs their employee maintenance, which includes user creation and assignment within the organization structure (role assignment) in their existing ERP system. The organization structure in C4C will need to be configured so that it is in sync with the organizational structure in the ERP system (Figure 33). Users will not be automatically created in C4C, they will require manual creation and role assignment.

Assignment Plan (CRM)	Code	ID	Business part...	Valid from	Valid to	Assigned a...	Assigned u...	Percent...
Customer Service	CusSrv	O 20005168	OU20005168	06/01/2008	Unlimited			
Executive Assistant 2	EXA2	S 50094088		06/17/2012	Unlimited	04/01/2013	Unlimited	
Customer Programs & Services	00390	O 20005169	OU20005169	06/01/2008	Unlimited	06/01/2008	Unlimited	
VP Customer Programs and Services	VP CusPrgrsSv	S 50016054		06/01/2008	Unlimited	06/02/2014	Unlimited	
Executive Assistant 2	EXA2	S 50031716		06/01/2008	Unlimited	03/06/2012	Unlimited	
Consumer Affairs	00674	O 20005277	OU20005277	06/01/2008	Unlimited	04/01/2013	Unlimited	
Product Management	ProdMgmt	O 20005344	OU20005344	06/01/2008	Unlimited	11/03/2014	Unlimited	
DSM Program Management	DSMPrgrMgmt	O 20007327	OU20007327	06/01/2008	Unlimited	04/01/2013	Unlimited	
Engineering Services	EngnSrvcs	O 20007329	OU20007329	06/01/2008	Unlimited	03/07/2016	Unlimited	
Product Development & Div Mgmt	ProdDev&DivMg	O 20020163	OU20020163	09/03/2014	Unlimited	09/03/2014	Unlimited	
Customer Experience & Insights	CustExpnce	O 20020860	OU20020860	04/06/2015	Unlimited	04/06/2015	Unlimited	
Business Customer Division	00395	O 20005171	OU20005171	06/01/2008	Unlimited	06/01/2008	Unlimited	
Customer Service Operations Division	CSOD	O 20005631	OU20005631	06/01/2008	Unlimited	06/01/2008	Unlimited	
VP Customer Service Operations	VP CusSrvOp	S 50020538		06/01/2008	Unlimited	08/08/2016	Unlimited	
Revenue Services Organization	00429	O 20005186	OU20005186	06/01/2008	Unlimited	06/01/2008	Unlimited	
Dir Revenue Services Organization	DirRSO	S 50101606		04/01/2013	Unlimited	04/01/2013	Unlimited	
Credit & Payment Services	Crdt&PymtSrv	O 20005188	OU20005188	06/01/2008	Unlimited	07/08/2013	Unlimited	
Manager-Project/Product 1	MPP1	S 50020111		06/01/2008	Unlimited	09/19/2016	Unlimited	
Dir Receivables Management	Dir RcvblsMg	S 50070157		07/19/2010	Unlimited	05/02/2016	Unlimited	
Administrative Assistant	AID4	S 50140032		01/17/2017	Unlimited	01/17/2017	Unlimited	
Administrative Assistant	AID4	S 50140936		03/02/2017	Unlimited	03/02/2017	Unlimited	
Collection Operations	ColOp	O 20005623	OU20005623	06/01/2008	Unlimited	06/01/2008	Unlimited	
Analyst-Business 3	ABU3	S 50067407		04/12/2010	Unlimited	03/23/2015	Unlimited	
Analyst-Business 3	ABU3	S 50096563		09/12/2012	Unlimited	03/23/2015	Unlimited	
Sr Mgr	ALL915_M3_E	S 50137887		10/07/2016	Unlimited	10/07/2016	Unlimited	
Customer Solutions Representative 2	CSR2	S 50144339		06/23/2017	Unlimited	06/23/2017	Unlimited	
Customer Solutions Representative 2	CSR2	S 50144360		06/23/2017	Unlimited	06/27/2017	Unlimited	
Customer Solutions Representative 2	CSR2	S 50144361		06/23/2017	Unlimited	06/23/2017	Unlimited	
Customer Solutions Representative 2	CSR2	S 50144463		06/26/2017	Unlimited	06/26/2017	Unlimited	
Analyst-Program/Project 3	APP3	S 50144547		07/06/2017	Unlimited	07/20/2017	Unlimited	
Active Collection	ActColl	O 20005625	OU20005625	06/01/2008	Unlimited	09/19/2016	Unlimited	
Customer Risk & Bankruptcy	CustRi	O 20006200	OU20006200	06/01/2008	Unlimited	06/01/2008	Unlimited	
Inactive Collection	Inactv	O 20006204	OU20006204	06/01/2008	Unlimited	03/23/2015	Unlimited	
Payment Management	PymtMg	O 20005627	OU20005627	06/01/2008	Unlimited	06/01/2008	Unlimited	
Credit Risk Mgmt Strategy	CreditRiskMg	O 20022266	OU20022266	09/19/2016	Unlimited	09/19/2016	Unlimited	
Prccs Mgmt & Stgy Integration	PrccsMgtStr	O 20005341	OU20005341	06/01/2008	Unlimited	07/28/2014	Unlimited	
Customer Billing Operations	CustBillOps	O 20005651	OU20005651	06/01/2008	Unlimited	06/01/2008	Unlimited	
Work Mgmt & Digital Labor	WkMgmt	O 20007213	OU20007213	06/01/2008	Unlimited	04/01/2013	Unlimited	
Oper Quality & Compliance	OpQua	O 20012455	OU20012455	03/22/2010	Unlimited	04/01/2013	Unlimited	
Customer Contact Center	CCC	O 20005187	OU20005187	06/01/2008	Unlimited	06/01/2008	Unlimited	
Safety & Business Organizational Mgmt	SafeBusOrgMg	O 20007885	OU20007885	07/21/2008	Unlimited	08/22/2016	Unlimited	
Prof Svcs Support & Integration	ProfSvcs	O 20014991	OU20014991	09/26/2011	Unlimited	01/10/2014	Unlimited	
Sourcing Management	SourMgmt	O 20019680	OU20019680	01/10/2014	Unlimited	01/10/2014	Unlimited	

Figure 33: SCE's Existing Organizational Structure

2.1.6.2 Organizational Structure and Roles - Roles

At the time of writing, the user groups have been identified and a preliminary assessment of their required access has been done to identify the roles that are required. At a high level, we've defined the following roles:

- Contact Center Agent Roles: This role will be used by all contact center agents to process inbound calls and process inquiries. This will provide user's access to C4C and S4/HANA through FOPs. The SAP recommended role "Utilities – Call Center Agent" will be leveraged and modified per SCE requirements.
- Specialty Operations: This role will be used by all employees within the CCC Specialty Operations department. Contact Center Supervisor Roles: This role will be used by contact center supervisors, which will have access to all contact center agent functions, as well as elevated access to certain functions including generating a manual invoice. "Utilities – Call Center Agent" will be leveraged and modified per SCE requirements.
- BCD Role: A role will be created for the BCD group which will be used by the following:
 - BCD AM's/IAM's & BCD Super Users
 - BCD Analysts
 - BCD Management-VP-M2's and M1's
 - Field Engineers/Pump Test man/ /DER Specialists
 - Charge Ready
 - Rate Analysis
 - Territory Management

The primary difference in the BCD role is in the ability to create credit options (installment plans, level pay plans etc.). Accounts that are managed by BCD do not qualify for these offerings, and therefore, the BCD group does not have access. BCD will process Move In's (SAP recommended roles are "Utilities – Key Account Manager" and "Sales Manager" which will be leveraged and modified per SCE requirements.

- CP&S Roles: Two roles will be created for the CP&S that will be used by the following groups:
 - CP&S Product Management super users and
 - CP&S Project Management and Analysts.
 - CP&S Digital Services

This user group is primarily responsible for managing programs, but will require limited access to C4C. The SAP recommended role is "Utilities – Call Center Agent" and "Sales Manager" which will be leveraged and modified per SCE requirements.

- RSO Role: The RSO group is a back-office group within CAPS, which will use C4C on a limited basis. A role has been defined for them that will provide them access to view account information in C4C. The SAP recommended role is "Utilities – Call Center Agent" which will be leveraged and modified per SCE requirements.
- IQP Role: A role has been defined for IQP. Their access is similar to CP&S, with a few minor differences.

The preliminary user groups and access requirements are detailed in the Business Roles document posted in the APSE and embedded below. This document only lists roles and access to C4C and does not include access to FOB to the back office, iEnergy, Hybris Sales, BDEX, etc.



C4C Roles For
Businesses 5.22.2018.

2.2 Process / Performance Improvement

- SAP allows for smart search capabilities and provides suggestions as the user types, similar to text prediction. It also provides a wildcard search option using an asterisk (*). Users can search by multiple criteria such as name and house/business number.
- Web-based, customer service, sales, and social networking functionality
- Provides an intuitive, Web-based user interface that is compatible with mobile devices.

2.3 Prerequisites and Inputs

- End-users have been created within both C4C and S4/HANA, and should be assigned their appropriate roles
- Master Data has been successfully loaded into both S4/HANA and Hybris C4C
- 3rd Party Integration is complete
- CPI connection is established successfully

2.4 General Assumptions

- Business Master Data and Technical Master Data will need to be loaded per applicable BPD
- Computer Telephony Integration (CTI) (as indicated in IVR BPD)
- Cloud Platform Integration (CPI) configuration should be complete

2.5 Business Process Hierarchy

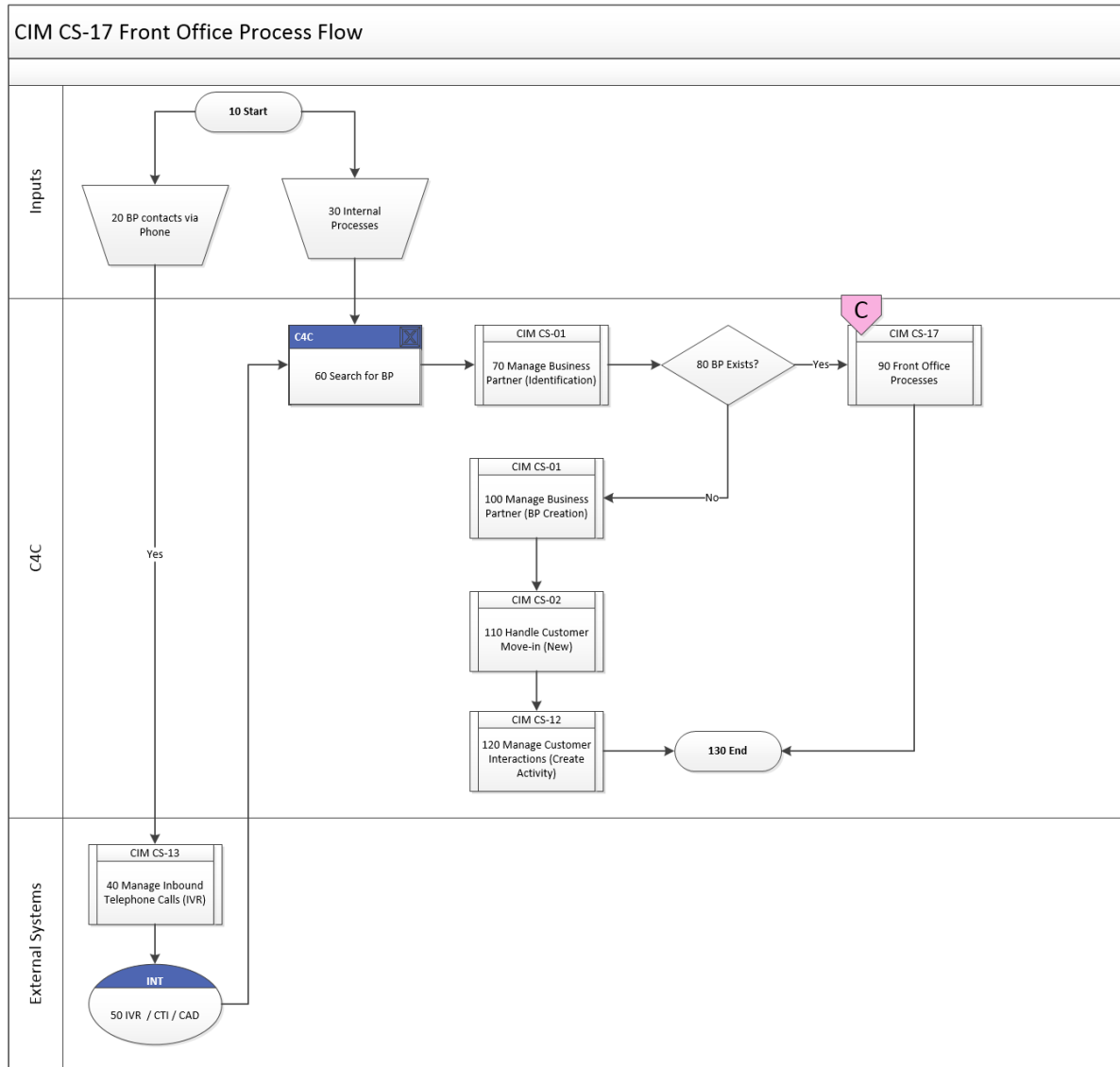
This Business Process Hierarchy provides a listing of all scenarios/business processes, sub-processes, and SAP Transaction codes identified to support this process.

Level 1	Level 2	Level 3	Level 4
Front Office and Back Office	Business Partner	Create Business Partner Change Business Partner Display Business Partner Create Contract Account Change Contract Account Display Contract Account Utilities 360 Utilities Financial Utilities Rate Change Utilities Exception Utilities Service Order Utilities Service Notification Utilities Sales POD Account Hierarchy	Front Office/C4C System

2.6 Process Flows

2.6.1 Process Flows

Process Flow Diagram



Step	Description of Step	C4C / T-Code	Role(s)	If step is relevant for a Control, add Requirement ID
10	Start	N/A	N/A	N/A
20	The customer contacts SCE via inbound phone call.	N/A	N/A	N/A
30	The customer may contact SCE via email or letter, or any other internal SCE process which requires a front office agent to review the account.	N/A	N/A	N/A

Step	Description of Step	C4C / T-Code	Role(s)	If step is relevant for a Control, add Requirement ID
40	The customer contacts via inbound phone, which initiates process CIM CS-13 Manage Inbound Phone Calls (CTI).	CTI	N/A	N/A
50	Integration with IVR to authenticate the customer and provide the call center agent the caller attached data. Details are documented in CIM CS-13 Manage Inbound Phone (CTI)	N/A	N/A	N/A
60	Search for the BP in C4C.	C4C	- o Contact Center Agent Role - o Contact Center Supervisor Role - o BCD Role - o RSO Role - o IQP Role	N/A
70	Business Partner identification is documented in CIM CS-01 Manage Business Partner.	C4C	- o Contact Center Agent Role - o Contact Center Supervisor Role - o BCD Role - o RSO Role - o IQP Role	N/A
80	Decision: Does the BP already exist? If the BP already exists, proceed to step 90. If the BP does not exist, proceed to step 100.	C4C	- o Contact Center Agent Role - o Contact Center Supervisor Role - o BCD Role - o RSO Role - o IQP Role	N/A

Step	Description of Step	C4C / T-Code	Role(s)	If step is relevant for a Control, add Requirement ID
90	Once the customer has been authenticated, this process will subsequently initiate any of the following processes: ○ CIM-CS-01 - Business Master Data - Manage Business Partner (BP updates) ○ CIM-CS-02 - Handle Customer Move-in/Out - Contract New ○ CIM-CS-03 - Handle Customer Move-in/Out - Contract End ○ CIM-CS-04 - Handle Trouble & Outage Contacts - Process Trouble Call ○ CIM-CS-05 - Handle Request - Process Field Work Request ○ CIM-CS-06 - Handle Customer Inquiries ○ CIM-CS-07 - Handle Bill Inquiries ○ CIM-CS-08 - Complaints, Claims, and Survey ○ CIM-CS-09 - Manage Inbound Request ○ CIM-CS-14 - Manage Owner Allocation ○ <u>PP-CS-02 - Program Enrollment and Delivery, De-Enrollment & Decommission Programs</u> ○ <u>PP-CS-03 - Fulfill Customer Request</u> ○ <u>PP-CS-04 - Manage & Support Program Operations</u>		○ Contact Center Agent Role ○ Contact Center Supervisor Role ○ BCD Role ○ RSO Role ○ CP&S Role ○ IQP Role	N/A

Step	Description of Step	C4C / T-Code	Role(s)	If step is relevant for a Control, add Requirement ID
100	BP Creation is documented in CIM CS-01 Manage Business Partner.	C4C	- Contact Center Agent Role - BCD Role	N/A
110	Creating the customer's contract (turn-on for start service) is documented in CIM CS-02 Handle Customer Move-in (New)	C4C	- Contact Center Agent Role - Contact Center Supervisor Role	N/A
120	Once the customer request is completed, the agent will need to create an activity to log the customer interaction. This is detailed in CIM CS-12 Manage Customer Interactions.	C4C	- Contact Center Agent Role - Contact Center Supervisor Role	
130	End of Process	N/A	N/A	

2.7 3rd Party and SCE Legacy System Integrations

3 rd Party or SCE Legacy System Integration	Description of Integration and Required Change to 3 rd Party or SCE Legacy System	Estimated Complexity of 3 rd Party or SCE Legacy system change
IVR	CTI Integration	

2.8 Data Elements

N/A

2.8.2 Source Systems and Data Volumes

The number of annual business transactions processed by the call center are specified below:

- 2015: 31,972,803 transactions per year
- 2016: 31,300,064 transactions per year
- 2017: 31,113,700 transactions per year
- Average: 31,462,189 transactions per year for the 3 years

2.9 Cloud for Customer (C4C) Integration

2.9.1 Processes

N/A

2.9.2 C4C Views

N/A

2.9.3 Interaction Records

N/A

2.10 Exception Management Requirements

This section outlines a high level description of each BPEM Case or other user workflow items (such as “Tickets” used by Front Office) used to route work requests between business areas. The companion document for this section which outlines the detail of each case is documented in APSE on the [Exception Case Template General Document #8](#).

User Exception Scenario / Description	User Group	Initiated by	Work Distribution
Note: Please refer to each individual applicable BPD to view exceptions for a specific process. For example, exceptions related to Move In are documented in BPD CIM-CS-02 Handle Customer Move-in Out - Contract New			

2.11 Process Risks and Mitigation

Risk Statement	Mitigation
C4C quarterly release	Hybris C4C is a cloud solution, which receives regular quarterly updates from SAP. While the updates are intended to release new functionality, the code delivered in the updates could implement changes, which could adversely impact RICEFW objects that were already developed, or cause program defects within the standard code. To mitigate this risk, the project team will closely monitor all quarterly releases for potential impacts.*

*Cloud controls framework will be defined at the enterprise level to ensure adequate design by the CM SMOO and EY.

2.12 Key Performance Indicators

The following have been identified as KPI’s at Southern California Edison, along with the means of measuring each KPI and the current measurement statistic for each KPI:

Measured KPI	Description	Measurement Method	Current Statistic
N/A			

2.12.1 Key Metric Definition

The following have been identified as key metrics at Southern California Edison that require business rule for calculation:

Key Metric	Description	Business Rule	Reference
N/A			

3 Controls, Regulatory Rules, and Security

Requirement ID (from RTM)	"As-Is" Control Requirement	"To Be" Risk Statement	"To-Be" Control Activity	Control Automation (manual, IT dependent-manual, automated)	Relevant technical identifier or RICEFW reference
CONTROL-CS-1199	Training On Communicating Accessible Authorized Payment Agency (APA) Locations (DisabRA Settlement Agreement): SCE is required to provide customer service training on communicating accessible Authorized Payment Agency (APA) locations to callers. (See DisabRA Settlement Agreement, III.B.2.f).	Noncompliance or misinterpretation with regulatory and legislative requirements may lead to misstatement of financial statements, penalties, fines and/or prosecution from governing regulatory bodies	On an annual basis, CCC management provides refresher training to all CCC Agent to provide Authorized Payment Agency (APA) location upon request of all customers, including accessibility information to the DisabRA customers. All new employees are provide this training as part of the New Hire Training.	Manual	N/A

3.1 Security Considerations

Role	Transaction Code (T-Code)	T-Code Description	Functional Restriction	Organizational Restriction
Please refer to the roles spreadsheet in section 2.1.6.2				

3.2 Custom Program/Transaction Codes

Please identify any custom programs/transaction codes that need to be developed for RICEFW objects. Include RICEFW Reference ID from APSE in RICEFW Reference ID column.

The following details should be identified and captured in their respective RICEFW Functional/Technical specs

- Is the custom program executed in the batch mode or dialogue mode
- Is the custom program assigned to an authorization group
- What is the custom transaction code for the custom program
- Should there be an authorization check made against any field in the custom program/custom t-code and if so what is the custom/standard authorization object used(For example does it have to be restricted to residential vs commercial customer)

- Does the custom programs/custom t-codes have any critical or sensitive customer information that needs extra security

Custom Table	Authorization Grp Assigned	Custom T-code for the Table	Table Access (display/update)	RICEFW Reference ID
N/A				

3.3 Custom Table/Transaction Codes

Please identify any custom tables/transaction codes that need to be developed for RICEFW objects.

Include RICEFW Reference ID from APSE in RICEFW Reference ID column.

The following details should be identified and captured in their respective RICEFW Functional/Technical specs

- Are there any customer sensitive/critical Z tables created
- What is the authorization group assigned to the table
- Is it just a maintenance or display only table
- What is the Z transaction code for the custom table

Custom Table	Authorization Grp Assigned	Custom T-code for the Table	Table Access (display/update)	RICEFW Reference ID
N/A				

3.4 Data Criticality and Cybersecurity Baseline Risks

Specify Data Classification /Special Handling or Regulatory Requirement <i>What is the classification of Data? Does it trigger any special regulations?</i> <i>Please select the classification / Sensitivity of the Data. This will help us Assess the Risk associated and remediation measures if any.</i>	<Select all that apply> ☒ PII - Personal Identifiable Information / Privacy Law ☐ CEII - Critical Energy Infrastructure Information ☐ FERC SOC (Standards of Conduct) ☐ NERC CIP Confidential ☐ Sarbanes-Oxley (SOX) financial reporting data ☐ SCE utility data accessible by affiliate companies (CPUC) ☐ PCI - Payment Card Industry Compliance ☒ Customer Information Service Request (CISR, CISR-DRP) ☐ Other <u><Please Specify></u> ☒ Internal ☐ Public ☒ Confidential
Please identify which business Group/Role will require access to the data and/or solution: <i>Please select the list of Business Groups will have access to this Data / Process. This will help us Determine the Impact and Risk</i>	<Select all that apply> ☐ SCE Customer, Res./Comm. Customer; ☒ SCE OU, Business Function Organization Name: CCC, BCD ☐ Vendor ☐ External business partner(s) & SaaS providers ☒ External on-shore contractors ☒ External off-shore contractors ☐ SCE Affiliate Company Employees

<i>Spread. Will ensure efficient Risk mitigation measures.</i>	☒ SCE Employees, specify if they have Marketing Function responsibilities (e.g., ES&M) ☐ SCE contract workers ☐ Regulatory Agencies ☐ Others <u><Please Specify></u>
How many of the SCE Customer, Employee, Vendors will use or affected by the Business Process? <i>To identify the Impact if services are adversely affected.</i>	1,000 - 10,000
How much Information/data would be processed, received or transmitted by the Business Process? <i>The data volume that will be utilized by this process. Helps us in determining the Impact and criticality</i>	
Number of business transactions in average? <i>These are the number of transactions process by the application on average. The number of transactions help in understanding criticality and Impact.</i>	2015: 31,972,803 2016: 31,300,064 2017: 31,113,700 Average: 31,462,189
Additional Security Consideration:	Any Cybersecurity impact in addition to SAP system security: • Internet Facing Web Application? Yes Details: C4C is a cloud based application. • Internet Facing Application Programming Interface? Yes. • 3rd party Federated Authentication & Authorization, Single Sign On? Yes.

4 Technical Considerations

4.1 RICEF-W Assessment

All of the Forms, Reports, Interfaces, Conversion items, Extensions/Enhancements and Workflows (RICEF-W) that have been identified for this BPD are detailed in the tables below and will represent the totality of all RICEF-W items documented at the time of sign-off of this BPD. Any RICEF-W items that are confirmed via change control after this BPD receives final sign-off, will be managed within Southern California Edison's APSE environment and not within this document.

Items denoted with ** have been identified as duplicate FRICEW objects.

3.1.1 Reports

Report Name & Description (List Requirements met by this Report)	Complexity	Reference ID
IR CIM CS 17 – Customer Hierarchy Report	Medium	1614

3.1.2 Interfaces

Interface Name & Description (List Requirements met by this Interface)	Complexity	Reference ID
I-CIM CS 17 – Receive outage response from BP	Medium	707
I-CIM CS 17 – Add Note	Medium	708
I-CIM CS 17 – Appointment Booking	Complex	1855
I-CIM CS 17 – View Appointment Times	Medium	710

3.1.3 Conversion

Conversion Name & Description (List Requirements met by this Conversion)	Complexity	Reference ID
C-CIM CS 17 – Conversion for Historical CCA Contracts	Medium	140
C-CIM CS 17 – Conversion for Rate	Medium	141
C-CIM-CS-17 - Conversion for Service Confirmations (Required for energy audits)	Medium	143

3.1.4 Extension/Enhancements

Extension Name & Description (List Requirements met by this Extension)	Complexity	Reference ID
E-CIM-CS-17 - Bolt-On	Medium	1803
E-CIM-CS-17 - CA Custom Fields Requirement - 1879 - CC-FI-05-R0005 Requirement - 8378 - CIM-CS-17-R0055 Requirement - 8381 - CIM-CS-17-R0058	Medium	1804
E-CIM-CS-17 –Custom View for IDEX This is to provide a view for CCA-DA customers	Complex	1062

Extension Name & Description (List Requirements met by this Extension)	Complexity	Reference ID
E-CIM-CS-17 - Consolidated Bill View Requirement - 8382 - CIM-CS-17-R0059 Requirement - 1920 - CC-FI-02-R0012	Complex	1530
E-CIM-CS-17 - Consolidated Communication View Requirement - 8385 - CIM-CS-17-R0062	Medium	1532
E-CIM-CS-17 - Consolidated Rate Analysis View Requirement - 8387 - CIM-CS-17-R0064	Medium	1534
E-CIM-CS-17 - Consolidated Trouble Orders View	Medium	1531
E-CIM-CS-17 - Credit Check Requirement - 595 - CIM-CS-02-R0045 Requirement - 618 - CIM-CS-02-R0068	Complex	1805
E-CIM-CS-17 - C4C Outage View Requirement - 7758 - CIM-CS-04-R0002 Requirement - 7792 - CIM-CS-04-R0036 Requirement - 7793 - CIM-CS-04-R0037 Requirement - 7801 - CIM-CS-04-R0045	Complex	1808
E-CIM-CS-17 - Supply Scenario Requirement - 642 - CIM-CS-02-R0092	Simple	1809
E-CIM-CS-17 BP Contacts in C4C	Simple	2017
E-CIM-CS-17 BPEM Creation in C4C	Simple	2021
E-CIM-CS-17 Create Service Orders in C4C	Simple	2019
E-CIM-CS-17 Rate Change Validations in C4C Requirement - 553 - CIM-CS-02-R0003	Medium	2022
E-CIM-CS-17 - Fiori Billing Overview	Complex	778
E-CIM-CS-17 - Move Out**	Complex	1540
E-CIM-CS-17 - Account Manager Notification**	Medium	1836
E-CIM-CS-17 - Activity Authorization**	Medium	1837
E-CIM-CS-17 - Activity Auto Populate**	Medium	1838
E-CIM-CS-17 - Activity Custom Fields**	Simple	1819
E-CIM-CS-17 - Activity Flow**	Simple	1839
E-CIM-CS-17 - Activity Search**	Simple	1840
E-CIM-CS-17 - Activity Update**	Simple	1841
E-CIM-CS-17 - Address Failure BPEM	Medium	1549
E-CIM-CS-17 - Alert (Recommended) Requirement - 7489 - CIM-CS-01-R0142 Requirement - 7491 - CIM-CS-01-R0144 Requirement - 8513 - CIM-CS-17-R0190	Complex	1496
E-CIM-CS-17 - Alert Handling**	Complex	779
E-CIM-CS-17 - Alert Handling**	Complex	1508

Extension Name & Description (List Requirements met by this Extension)	Complexity	Reference ID
E-CIM-CS-17 – Appointment scheduling	Medium	1535
E-CIM-CS-17 – BBP Validations**	Medium	1842
E-CIM-CS-17 – Bill Protection calculation**	Medium	1539
E-CIM-CS-17 – BP Account Summary (Desired)	Medium	1843
E-CIM-CS-17 – BP Search (Desired)	Medium	1846
E-CIM-CS-17 – BPEM to create dunning lock for Bill Dispute	Medium	1544
E-CIM-CS-17 – Communication to BP to verify contact information	Complex	1543
E-CIM-CS-17 – Configurable Hierarchy**	Medium	1847
E-CIM-CS-17 – Confirmation to customer on inbound communication received**	Complex	1541
E-CIM-CS-17 – Consolidated Rates View Requirement - 8386 - CIM-CS-17-R0063	Complex	1533
E-CIM-CS-17 – Consumption Graph	Complex	1820
E-CIM-CS-17 – CRM Outage History**	Complex	795
E-CIM-CS-17 – Custom Search Criteria**	Medium	803
E-CIM-CS-17 – Customized CRM_IC – Account Overview**	Complex	785
E-CIM-CS-17 – Default Streetlight rate for Metered Streetlight line item**	Complex	1536
E-CIM-CS-17 – Device Location Notes (Desired)	Simple	1848
E-CIM-CS-17 – Dunning Next Step (Desired)	Medium	1849
E-CIM-CS-17 – Field Work Request**	Medium	1806
E-CIM-CS-17 – Joint Invoice Change (Recommended) Requirement - 2885 - UB-BI-10-R0007	Simple	1807
E-CIM-CS-17 – Linkage for parent and child BPEM (Recommended)	Medium	1545
E-CIM-CS-17 – NEM Rate**	Medium	1538
E-CIM-CS-17 – Next Bill date** Requirement - 576 - CIM-CS-02-R0026 Requirement - 577 - CIM-CS-02-R0027	Simple	1850
E-CIM-CS-17 – Outbound communication status screen	Medium	1542
E-CIM-CS-17 – PnP Enrollment (Recommended)	Complex	1546
E-CIM-CS-17 – PnP Enrollment Eligibility (Recommended)	Complex	1548
E-CIM-CS-17 – PnP Enrollment Status (Recommended)	Medium	1547
E-CIM-CS-17 – PremiseOV (Recommended)	Medium	1529

Extension Name & Description (List Requirements met by this Extension)	Complexity	Reference ID
E-CIM-CS-17 – ServiceOrder History (Desired)	Complex	796

3.1.5 Forms

Form Name & Description (List Requirements met by this Form)	Complexity	Reference ID
None.		

3.1.6 Workflows

Workflow Name & Description (List Requirements met by this Workflow)	Complexity	Reference ID
None.		

3.2 IDEX Considerations

An enhancement has been identified for IDEX: [1062 - E-CIM-CS-17 - CCA DA Verification](#)

3.3 Production System Cut-over Considerations

- Hybris C4C connectivity to S4/HANA will need to be established as a part of cutover. This includes the creation of the communication systems, communication arrangements, and configuration and deployment of Iflows within CPI.

4.6 Other Considerations

N/A

4 Change Management

5.1 Change Impacts

Change Element Short Description	Change Impact Category	Audience Group(s)	Impact	Business Effort
All transactions/orders must replicate to the backend from the front end tool. If it fails, someone must address the replication error. A new role will be required to address exceptions. SCE business users are more appropriate to work on replication issues as SCE IT may not have the authorized system functionalities to fix the data compared to business users	Job Roles (35)	CS-CSOD-CCC All, CS-CSOD-RSO All, CS-BCD-All	Low	Low
SAP releases new value-added features and upgrades for the C4C application on a pre-planned schedule. These releases will need to be tested. New features from the release will need to be communicated to users as well.	Technology	CS-CSOD-CCC All, CS-CSOD-RSO All, CS-BCD-All	High	High
SAP has different terminology between the Front office and Back office applications. (Example - Business Partner in Back-Office and Customer in Front Office)	Process (122)	CS-CSOD-RSO All; CS-CSOD-CCC All; CS-BCD-All; CS-CSOD-Consumer Affairs	Low	Low
Currently customer account numbers are 9 digits in length and begin with the number 2. In SAP, Contract Account numbers will be 12 digits in length and the leading indicator will be a 4, 5, 6 or 7 depending on the account type. An account number beginning with a 4 will be a 3rd Party Supplier account, a 5 will be a stand-alone Deposit account, a 6 will be a Collective account, a 7 will be a Standard or Joint Contract. Users will be able to locate the legacy account number in the BP level. Users will need to advise customers about the new account identifiers and where to locate it on their statement in the event that they call the contact center and use their old CSS account number as a search criteria. External customers will be impacted and can drive call volume. For back office users, it may add additional AHT for account look up in the MDMS and will have an impact to CCA/DA third parties. It may also have reporting impacts as well as impacts to external vendors. Customer Impact - Customers will need to locate their new account numbers on their bill statement as well as make any changes to their accounts that require the account number such as banking institutions, reporting needs etc.	Process;Decision Making (127)	CS-CSOD-CCC All; CS-CSOD-RSO All; Ext-Customers-All; Ext-Third Party Vendors-All; Ext-Third Party Vendors-Banks; CS-BCD-All; CS-CP&S-All; CS-CSOD-Consumer Affairs	Medium	Medium

Change Element Short Description	Change Impact Category	Audience Group(s)	Impact	Business Effort
Mailing address verification process is currently performed by third-party and this will be changed in SAP (Address verification will be done in SAP). Going from 1 month to 4 months. Concern – Increase of return mail. We need to ensure this will not impact the cost savings with USPS and Compliance. May increase operational work to handle increase of return mail and manual clean-up of addresses. It may have O&M budget implications for RSO mailing operations	Process;Regulatory/Legal;Technology (132)	CS-CSOD-CCC All;CS-CSOD-RSO All;CS-BCD-All;Compliance-All;CS-CP&S-All;CS-CSOD-Consumer Affairs	Medium	Low
SAP allows for smart search capabilities. It will provide suggestions as the agent types in more data. It can also provide a wildcard search option (using *). It also allows to search by multiple criteria such as name and house number. CSS currently does not have the functionality. The agent has to type the street name, customer name or account number as search options. As a result, searches should be quicker and provide more flexibility.	Process;Benefit/Improvement (208)	CS-CSOD-CCC All;CS-BCD-All;CS-CSOD-RSO All;CS-CSOD-Consumer Affairs	Low	Low
In SAP, there are multiple functions related to tracking of customer interactions. Users can create activity records with notes attached or an interactive free form notes field. Other users will create notes attached to BPEM cases and BP contacts notes. Currently there is only 1 notes function in CSS from the different levels. Users will need to know the difference on each of these and know when to use the proper notes or record to ensure consistency in the process	Process (241)	CS-CSOD-CCC All;CS-CSOD-RSO All;CS-CSOD-Consumer Affairs	Low	Low
Average handle time may increase in the front office due to multiple changes in the processes. Front line users will have the capability to complete more transactions from an end to end perspective as well as certain processes have more steps than as is process. Both of these can increase AHT. The ability to complete more transactions will improve customer satisfaction as well as first call resolution if agents have the ability to issue and correct more issues while they are interacting with the customer but AHT impacts associated with those calls types will need to be reviewed and reassessed.	Compensation/Performance;Culture Shift;Process;Staffing Levels(368)	CS-CSOD-CCC All;	High	High
Pending SAP Upgrade 1808 - Front office users will have visibility to single BPEM case through the life cycle of the work item. This can be used for SLA compliance or advise customer of the detailed status of the request for customer satisfaction and first call resolution. Currently users do not have detailed visibility to the life cycle of the exception.	Benefit/Improvement;Decision Making;Process;Regulatory/Legal;Compensation/Performance (496)	CS-BCD-All;CS-CP&S-All;CS-CSOD-All	High	High

Change Element Short Description	Change Impact Category	Audience Group(s)	Impact	Business Effort
DSM and Energy Savings programs process may change based on the new bolt on process integration. The lead/opportunity process will change as a result of outputs from the bolt on system. SAP front office User Interface will have an integration with the new bolt on systems. Currently DSM opportunities are created and tracked in the Legacy CRM system.	Culture Shift;Process;Technology;Benefit/Improvement (517)	CS-BCD-All;CS-CSOD-CCC-PS;CS-CSOD-CCC-EE	High	Medium
Authorized users (e.g. supervisors and managers) need to be trained on the process to ensure that roles are transferred properly/correctly if moving from one position to another position (onboarding). This needs to be trained so that employees maintain user productivity.	Decision Making;Job Roles;Process (521)	CS-All;T&D-All	High	High
The current system defaults state for drivers license to CA. In the new system agents will need to manually enter state for drivers license.	Process (952)	CS-CSOD-CCC All,CS-BCD-All	Low	Low
Currently there is a profile to indicate a Business Partner is a minor and an end date for when the person turns 18. This helps identify if the customer can be used for services In the future, the minor designation will need entered in Activities at the BP level. This information has the potential to be lost to the depth of notes in activities. Agents will need to be trained to search for this information.	Process, Decision Making (953)	CS-CSOD-CCC All;CS-CP&S-All;CS-CSOD-Consumer Affairs;CS-CSOD-RSO-Credit	Low	Low
If the BP resides in the city, CCA (Community Aggregate) is an automatic designation upon move-in. Currently the CCA opt out information is in the customer profile viewable by FO agents. In the future, agent will use FOP to access supply scenario on installation fact to view opt out information.	Process; Decision Making (954)	CS-CSOD-CCC All;CS-BCD-All;CS-CSOD-RSO All	Low	Low
Today a cut in flat rate is posted in notes. In the new system, a cut in flat identifier will be an installation fact. Pending requirements, a BPEM case will automatically be issued after a move in is submitted in a premise that is designated a cut in flat.	Process;Decision Making (956)	CS-CSOD-CCC All;CS-BCD-All;CS-CSOD-RSO All	Low	Low
Currently agents can add SSN and Tax ID in the FO screens during the move-in process. In the future, agents will create the BP, then FOP to enter SSN/Tax ID, execute the move-in.	Process (968)	CS-CSOD-CCC All;CS-BCD-All;CS-CSOD-RSO All	Medium	Medium
Today users view a change history log in CWO in view log and event history log. In C4C, users will need to view change history in service orders through FOP. Users will see the history of the system including the users who issued/updated and canceled the work order including any notes for a work order	Process;Decision Making (957)	CS-CSOD-CCC All;CS-BCD-All;CS-CSOD-RSO All	Low	Low
Legacy system provides all technical master data viewable to users. In the future there will be some technical master data in front office. If needed data is not visible the user will view through FOP to back office.	Process;Decision Making (974)	CS-CSOD-CCC All;CS-BCD-All;CS-CP&S-All	Low	Low

Change Element Short Description	Change Impact Category	Audience Group(s)	Impact	Business Effort
Front office users will access back office screens using the Front Office Process (FOP) launcher. It is a pending enhancement. The launcher creates an opportunity for users to view data and issue transactions that is not available in the front office UI (eg. installation facts, interval usage, utility tax exemption, joint billing etc.) and minimizes the need for a hand off to the back office. This creates a single sign on to the back office system from the front office. Currently CCC and BCD use one major system for most transactions, CWO and Legacy CRM respectively. Users will need to know when to use the FOP Launcher for transactions and data elements.	Culture Shift;Benefit/Improvement;Process;Decision Making (815)	CS-BCD-All;CS-CP&S-All;CS-CSOD-CCC All;CS-CSOD-Consumer Affairs	High	High
Mash up will be created to access multiple tools that the front office users will need to respond to customers' request or to complete a transaction (eg SPOC, iEnergy, GridX and Design Manager). A mash up allows the user to easily launch a 3rd party application via a link or by embedding it within C4C. Currently the primary system that CCC/GCS uses is CWO. BCD currently uses CSS and Legacy CRM for most of their customer needs. Additional training will be required for some of these systems that will be new to our agents	Decision Making;Process;Technology (1117)	CS-BCD-All;CS-CSOD-CCC All;CS-CSOD-Consumer Affairs	High	High
Currently, after completing a turn on request through SCE.com, they can also request to add the account to an existing summary billing arrangement. In SAP, SCE.com turn ons will only allow to set up as a solo account and not as part of a Joint or Collective invoicing arrangement. Operational Impact: Additional calls from the customer to complete their request where these were previously not received Customer Impact: Customers will be required to contact the CCC to add their new turn on account to a Joint/Collective arrangement	Decision Making; Process; Technology; Job Roles (Negative) 1043	CS-CSOD-RSO All;CS-CSOD-CCC All;CS-BCD-All;Ext-Customers-All	Medium	Medium
Currently users can place the turn on on hold by clicking certain check boxes that require the turn on to be held. For example, Same Party in check box, Remove meter/service checkbox, fuse up inspection, pos id investigation, no ssn information. In SAP, to place the move in "on hold" users will need to cancel the field work order and place a billing lock on the account to prevent the billing of services. Users will need to be trained on the process to place the move in on hold and how to reverse the process so that the move in can be completed.	Process (1041)	CS-BCD-All;CS-CSOD-CCC All;CS-CSOD-Consumer Affairs;CS-CSOD-RSO All;T&D-All	Medium	Medium

Change Element Short Description	Change Impact Category	Audience Group(s)	Impact	Business Effort
Users will access Design Manager through a Mash up to create New Business service requests (SR). There is no SR/MSR process in SAP so access to DM is necessary to begin the SR/MSR process. Currently users create an SR through CSS/CWO application. It allows users to create a New Business turn on along with the SR. Customer Impact: Customers will need to call multiple times to begin the SR process and then call again once the technical master data has been created to start the move in process. It may take multiple calls for the customer to complete the process. Service level agreements may be needed to discuss when the TMD is created so that customer facing users can provide accurate information to the customer.	Decision Making;Process;Technology (1130)	CS-BCD-All;CS-CSOD-CCC All;CS-CSOD-Consumer Affairs;CS-CSOD-RSO-CBO; Ext-Customers-All;T&D-DBL-Local Planning All;T&D-DBL-Tech Planning All;T&D-DBL-Track Planning All	Medium	High

Below is a description of each:

Change Element Short Description: Describes the ‘As-Is’ scenario and what is changing.

Change Impact Category: Determined by asking the questions below.

1. **Staffing Levels** - Is the work effort increasing or decreasing (staffing levels), and how (e.g., task automation, new process/tasks)?
2. **Job Roles** - Will the responsibilities, authority, or required skills for a job or job roles change? Will people be reporting to new leaders? Will managers’ responsibilities change?
3. **Compensation / Performance** - Will the changes have such an impact on roles of an individual that merits a change in how they are compensated or how their performance will be evaluated? What behaviors are rewarded and recognized?
4. **Organization Structure** - Is the ownership of the Process or Data changing (think departments, jobs, or roles, not individuals)?
5. **Decision Making** - Will any of the changes impact how decisions are made, by or between, business units / departments? Who makes them, how they determine the criteria for making a change?
6. **Culture Shift** - Will organizational value systems impact the way change happens or be impacted by the change? What are the underlying values of the organization that will directly influence the way change will be accepted and how much work will ultimately be required to ensure successful outcomes for the business?
7. **Location / Facilities** - Will the change impact the physical location where they process is being implemented or require changes to the facilities to enable the process to be effective?
8. **Process** - Any change to the current process (e.g. manual process becomes automatic, or vice versa)? Are there new processes being introduced or an old process being decommissioned? Is there a requirement to change an existing business rule or business policy? Will new business rule or policy needs to be introduced and enforced or complied to?
9. **Regulatory/ Legal** - Are there any legal / regulatory requirements that might be impacted
10. **Technology** - Are other applications impacted (e.g., going away, becoming read-only), or are there new requirements (e.g., new SW / HW, new version)
11. **Benefit/Improvement** - Will there be an improvement or benefit to any of the stakeholders because of this change. (Be sure to include any pain points that are being alleviated)
12. **Other** - Are there emotional implications stemming from any of these changes (think of buy-in issues or fears)

Audience Group(s): These are the affected departments, divisions etc...

Impact: The following were used to determine the impact:

Level	
HIGH 	- BUSINESS OUTCOMES: Drives change to cost, efficiency, customers. - JOB: More than 40% of individual's job will change and / or additional skills are required with significant changes to Job Documents. Examples include: - Shift from manually intensive role to using new tools and systems - Employee is going to be redeployed - High level of "technical or functional" changes needed for a particular group - Significant changes to culture & attitudes to get job done - Big shifts in accountability or reporting (autonomy, responsibility, decision making, etc.) - RELATIONSHIPS: High level of "relational" changes for a particular group (e.g. reporting relationships, effective working relationships with customers, teams, and stakeholders etc.) - POLICIES/PROCEDURES: Processes, procedures, and/ or policies are not in place today and need to be defined. - REGULATORY REQUIREMENT: drive changes to the way we comply
Level	
MEDIUM 	- BUSINESS OUTCOMES: Limited change to cost, efficiency, customers. - JOB: 20-40% of individual's job will change and / or additional skills are required with significant changes to Job Documents, Examples include: - Employee job title or function is modified, resulting in minor to moderate changes to roles and responsibilities. - There will be some changes to the culture & attitudes - Shift from moderate technology use to using new tools and systems - Moderate level of "technical or functional" changes needed for a particular group - Moderate shifts in accountability or reporting (autonomy, responsibility, decision making, etc.) - RELATIONSHIPS: Moderate level of "relational" changes for a particular group (e.g. reporting relationships) - POLICIES/PROCEDURES: Processes, procedures, and/or policies are inconsistent today and not defined or documented in many cases - REGULATORY REQUIREMENT: No impact.
Level	
LOW 	- BUSINESS OUTCOMES: No impact. - JOB: Less than 20% of individual's job will change and / or additional skills are required with significant changes to Job Documents. Examples include: - Employee job title or function is slightly modified with few, if any, changes to roles and responsibilities - There will be little change to the culture & attitudes - Shift from using some technology to new systems & tools - Light level of "technical or functional" changes needed for a particular group - Small shifts in accountability or reporting (autonomy, responsibility, decision making, etc.) - RELATIONSHIPS: No impact or light level of "relational" changes for a particular group (e.g. reporting relationships) - POLICIES/PROCEDURES: Processes, procedures, and/or policies are consistent, but not well documented, and will need to be documented/ Other minor changes to be made. - REGULATORY REQUIREMENT: No impact.

Business Effort: Initial assessment of the impact to the business High, Medium, Low

6 Requirements



CIM-CS-17 Design
Phase RTM (4).xlsx



Process Owner
Approval Document C